Technology in Society 82 (2025) 102900

Contents lists available at ScienceDirect

Technology in Society

journal homepage: www.elsevier.com/locate/techsoc

Can ex ante conformity assessment regulations contribute to trustworthy foundation models? An evolutionary game analysis from an innovation ecosystem perspective

Xiaoxu Zhang [a], Wenyong Zhou [a], Wen Hu [b], Shenghan Zhou [c], Xiaoqian Hu [d], Linchao Yang [e,*]

[a] School of Economics and Management, Tongji University, NO.1239 Siping Road, 200092, Shanghai, PR China
[b] Information Research Institute, Shanghai Academy of Social Sciences, No.1610 West Zhongshan Road, 200235, Shanghai, PR China
[c] School of Reliability and Systems Engineering, Beihang University, No.37 Xueyuan Road, 100191, Beijing, PR China
[d] School of Management and Engineering, Capital University of Economics and Business, No. 121 Zhangjialukou, 100070, Beijing, PR China
[e] School of Economics and Management, North China Electric Power University, No.2 Beinong Road, 102206, Beijing, PR China

ARTICLE INFO

Keywords:
Foundation models
Trustworthiness
Ex ante conformity assessment
Innovation ecosystems

ABSTRACT

Untrustworthy artificial intelligence (AI) systems, especially foundation models, may lead to significant economic and social issues, and this has been arousing widespread concern. However, there is no mature and future-proofed regulatory approach to govern foundation models or any consensus regarding the regulations due to their rapid development and limited understanding of them. Thus, the potential of alternative regulation methods should be fully discussed. The ex ante conformity assessment in the EU AI Act, the world's first comprehensive AI law, is applied to regulate high-risk AI systems and can be an alternative regulatory approach to manage foundation models in the future. Consequently, this necessitates considering whether ex ante conformity assessment can contribute to achieving trustworthy foundation models. Hence, we adopted an innovation ecosystem perspective and employed an evolutionary game approach, constructing two hypothetical scenarios for ex ante conformity assessment, namely, self-assessment and independent assessment. Findings show that market forces and ecosystem impacts play a crucial role in shaping trustworthiness and that ex ante conformity assessment alone—whether through self-assessment or independent assessment—may be insufficient to ensure trustworthy outcomes. Thus, we argue that market-driven incentives and ecosystem thinking among industry practitioners are pivotal for advancing trustworthy foundation models; however, it is important to remain cautious about the limitations of market mechanisms. Therefore, a hybrid regulatory framework that combines legal mandates with market-based incentives and ecosystem influences warrants further exploration. Furthermore, independent evaluators can serve as important facilitators in supporting providers through trustworthy audits. This study contributes to on-going policy discussions on trustworthy AI regulation and offers references for future policy design and implementation.

1. Introduction

Large language models have demonstrated impressive powers in understanding and generating natural language in recent years. These models, such as OpenAI's GPT models, Google's PaLM and Meta's LLaMa, are becoming increasingly influential with each iteration. A new term, 'foundation models', has been coined to refer to this class of generality-featured artificial intelligence (AI) systems. They are driving a paradigm shift as general-purpose AI and may profoundly affect various industries and the wider economy (Bommasani et al., 2022). However, the ascent of AI, especially foundation models, poses significant legal and ethical issues and new concerns for the future of work (European Commission, 2019a). Therefore, the idea of trustworthy AI has been gradually developed and is now widely acknowledged (Brundage et al., 2020; Floridi, 2019; Kaur et al., 2022). These worries are exacerbated by the fact that foundation models have unique

* Corresponding author.
E-mail addresses: zhangxiaoxu1993@126.com (X. Zhang), zhouwyk@126.com (W. Zhou), waitingforfox@163.com (W. Hu), zhoush@buaa.edu.cn (S. Zhou), huxiaoqian@cueb.edu.cn (X. Hu), yanglinchao@ncepu.edu.cn (L. Yang).

https://doi.org/10.1016/j.techsoc.2025.102900
Received 30 May 2024; Received in revised form 5 April 2025; Accepted 6 April 2025
Available online 9 April 2025
0160-791X/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

characteristics that could lead to various moral and societal hazards such as reinforcing social bias, breaching privacy, providing misleading information and causing economic loss due to system failure (Bommasani et al., 2022; Küspert et al., 2023; Li, Qi, et al., 2023; Papernot et al., 2017; Tramèr et al., 2016, pp. 601–618). Foundation models are evolving at an incredible pace, and their dramatically increasing capabilities and autonomy may soon amplify their impact massively, further increasing these risks and potentially resulting in large-scale social harm (Bengio et al., 2024).

Trustworthy AI is given significant attention by the European Commission (European Commission, 2019a, 2019b, 2020; The High, 2020). A pivotal step in this approach is the introduction of the world's first comprehensive AI law, known as the AI Act (European Commission, 2024; European Parliament, 2023), which identifies foundation models, also referred to as general-purpose AI models, as the central focus of governance (European Commission, 2024). However, despite the grave social concerns about the trustworthiness issues of foundation models, the stringent regulations applied to high-risk AI systems—whose trustworthiness is also a global concern—ex ante conformity assessment, encompassing both self-assessment and independent assessment, are not yet extended to foundation models (European Commission, 2024; European Union, 2024). The decision not to implement ex ante conformity assessment to foundation models seems a temporary measure, as internal evaluations, serving as a form of light-touch regulation, are relied upon to assess and mitigate potential risks and harms in these models (European Commission, 2024). This reliance is prompted by a current lack of expertise in conformity assessment and the on-going development of third-party auditing methods. While the development of deliberate policies, similar to those for high-risk AI systems, takes a considerable amount of time, the rapid development of foundation models emphasises the urgent need for the formulation of future-proof policies (Küspert et al., 2023). However, there is a lack of consensus about how to manage these models (Bengio et al., 2024). In addition, regulating an innovation process requires a thorough understanding of the potential effects of alternative regulations (Han et al., 2022). Consequently, fully exploring potential regulatory approaches that could prove effective in the future is imperative for establishing future-proof regulations. Thus, we attempt to analyse whether the regulations on high-risk AI systems could be applied to foundation models, as these regulations have the potential to improve the trustworthiness of foundation models in the future. In light of this, careful consideration should be given to several key questions related to the regulation of foundation models. First, is ex ante conformity assessment likely to be effective? Second, would conformity assessment methods—self-assessment or independent assessment—be more successful? Third, can trustworthiness be ensured by internal evaluations in the absence of market surveillance?

We employ an innovation ecosystem perspective to consider the influences of foundation models on other actors when discussing these questions. It is necessary to understand trustworthy foundation models from an innovation ecosystem perspective at a macro level rather than solely from a value chain or technology perspective. This is because, on the one hand, the profound influence of these models suggests that other actors in innovation ecosystems share a common destiny with foundation models and that they share the benefits and risks related to their trustworthiness at the ecosystem level. On the other hand, improving the trustworthiness of foundation models and fostering the responsible development of ecosystems require the joint participation of all actors involved. Some studies explore AI regulations from an ecosystem perspective (Knott et al., 2024; Nannini et al., 2024; Powers et al., 2023); however, the governance of foundation models within this framework requires further in-depth discussion. Considering the potent generative capabilities of foundation models, driving the development of diverse downstream models, applications and agents while fostering the emergence of ecosystems, their technological nature aligns with an ecosystem-oriented perspective (Bommasani et al., 2022; Küspert et al., 2023). Hence, the perspective of innovation ecosystems is crucial and warrants consideration. The concept of the innovation ecosystem is pivotal in comprehending the broader impacts of foundation models on the system overall, transcending a purely technical perspective and embracing innovation management considerations. We apply this perspective to explore trustworthiness issues and the corresponding regulations associated with foundation models.

We employ an evolutionary game approach to anticipate potential outcomes of ex ante conformity assessment and market surveillance as well as identify factors influencing the development of trustworthy foundation models. This approach is commonly used to analyse interactions among diverse participant groups within supervisory frameworks (Encarnaçao et al., 2018; Fan et al., 2021; Ji et al., 2019). Several studies on AI policies typically focus on regulatory frameworks based on trustworthiness principles and technological features; however, the strategic interactions between different actors require further exploration (Avin et al., 2021; Floridi, 2019). Therefore, applying an evolutionary game approach can complement existing research by offering deeper insights into these interactions.

In this study, our purpose is to scrutinise the potential contributions of ex ante conformity assessment and associated market surveillance to the development of trustworthy foundation models in terms of offering insights for achieving trustworthiness. We draw inspiration from the regulatory framework in the AI Act for high-risk AI systems and consider two policy scenarios: self-assessment and independent assessment (European Commission, 2024). Incorporating an innovation ecosystem perspective and leveraging an evolutionary game approach, we construct our analytical framework. Consequently, our findings indicate that provider-related and ecosystem-related factors may make a difference, and the development of trustworthy foundation models is not impacted significantly by ex ante conformity assessment, with market surveillance having a limited effect. Finally, we formulate conclusions and discuss the implications of our results. This study makes three primary contributions. First, it enriches innovation ecosystem research by applying this framework to understand the emerging AI paradigm of foundation models, while enhancing the understanding of foundation model ecosystems beyond a purely technological viewpoint, incorporating a valuable management perspective. Second, it further complements existing research on foundation model regulations by examining the interactions between different actors using an evolutionary approach. Third, it provides new insights into the regulation and advancement related to trustworthy foundation models for decision-makers in both industry and the public sector.

This study is organised as follows. We examine the literature on foundation models, AI trustworthiness, relevant legislation, innovation ecosystems and evolutionary game theory in Section 2. We build two evolutionary models in Section 3 based on the regulatory situations provided. Sections 4 and 5 describe model and numerical analyses, respectively. In Section 6, we offer our conclusions after discussing the implications with an eye to trustworthy foundation models from the viewpoints of policymakers and industry practitioners.

2. Literature review

2.1. Foundation models and trustworthiness

The term 'foundation model', coined by the Stanford Institute for Human-Centered Artificial Intelligence, is defined as 'any model that is trained on broad data (generally using self-supervision at scale) that can be adapted (e.g. fine-tuned) to a wide range of downstream tasks' (Bommasani et al., 2022). Although the technologies underpinning foundation models based on deep neural networks and self-supervised learning have existed for decades, they represent a new paradigm in AI fields, embodying artificial general intelligence (AGI) with generative capabilities (Bommasani et al., 2022). The AI landscape categorises technologies into artificial narrow intelligence and AGI (Fei et al.,

2022), with the former, or weak AI, 'trained on well-labelled datasets to perform specific tasks and operate within a predefined environment', and the latter, or strong AI, 'designed to perform a wide range of intelligent tasks, think abstractly and adapt to new situations' (Madiega, 2023). Moreover, foundation models demonstrate the ability to fuse data from various modalities and execute downstream tasks across multiple modes, including text, image, graph and other data modalities (Li, Tian, et al., 2023; Zhou et al., 2024).

Although the discussion on trustworthy foundation models remains limited, they fall under the broader umbrella of trustworthy AI, an area that has garnered significant attention in recent years. The trustworthiness of AI systems has been extensively deliberated across academia (Avin et al., 2021; Díaz-Rodríguez et al., 2023; Liu et al., 2022), the public sector (European Commission, 2019b; OECD, 2019) and industrial communities (Brundage et al., 2020; Cammarota et al., 2020). The European Commission, emphasising early AI governance, posits that establishing trust in AI is essential to unlock the full potential of these technologies (European Commission, 2019b; Laux et al., 2024; Thiebes et al., 2021). Drawing on this idea, it suggests three pillars for trustworthy AI, namely, the lawfulness, ethics and robustness of trustworthy AI (European Commission, 2019b). These serve as the basis for formulating the essential requirements (Díaz-Rodríguez et al., 2023), including (1) human agency and oversight, (2) technical robustness and safety, (3) privacy and data governance, (4) transparency, (5) diversity, non-discrimination and fairness, (6) societal and environmental well-being and (7) accountability (European Commission, 2019b). Subsequently, various essentials of trustworthy AI have been explored. Li et al. summarised the representative principles into technical requirements (robustness, explainability, transparency, reproducibility and generalisability) and ethical requirements (fairness, privacy and accountability) (Li, Qi, et al., 2023). Moreover, each of these principles should be considered throughout the entire lifecycle of the development, deployment and use of AI systems (Díaz-Rodríguez et al., 2023; Li, Qi, et al., 2023).

2.2. Regulations aiming for trustworthy foundation models

Due to the significance of trustworthiness, AI regulations have always been a topic of discussion. For example, Bengio et al. outlined a comprehensive plan that combines technical research and development with proactive, adaptive governance mechanisms (Bengio et al., 2024). Floridi, Veale & Borgesius, and Laux, Wachter & Mittelstadt analysed the important role of the EU regulatory approach and also point out its imperfections, which hinder achieving trustworthy AI (Floridi, 2019; Laux et al., 2024; Veale & Borgesius, 2021). Han et al. proposed a regulatory approach that combines a voluntary commitment approach reminiscent of soft law and analysed how it generates socially beneficial outcomes (Han et al., 2022). Stix proposed a framework based on the EU's AI Ethics Principles, aiming to increase practical implementability in policy processes (Stix, 2021). Avin et al. highlighted the effects of incident sharing, auditing and other concrete mechanisms on trustworthiness verification (Avin et al., 2021). Alalawi et al. analysed the possible effects of different regulatory regimes based on the interactions among users, AI creators and regulators (Alalawi et al., 2024). Bova et al. examined how governments can improve regulatory effects by purchasing services from a market of private regulators (Bova et al., 2023). Han et al. investigated the influence of rewards and punishments on the safe development of AI systems (Han et al., 2021).

Considering the ethical and social risks associated with foundation models, such as perpetuating stereotypes and social biases, providing false or misleading information and causing a range of human–computer interaction harms (Madiega, 2023), effectively governing these rapidly evolving and powerful models is an urgent priority. However, due to their fast-paced development and the limited understanding of their full impact (Bommasani et al., 2022), it is crucial to examine potential regulatory approaches from multiple perspectives to establish suitable and future-proof governance frameworks.

Various studies offer valuable insights into regulating foundation models. For example, Hacker et al. delineated the regulatory duties of various actors along the value chain of foundation models and proposed strategies for achieving trustworthiness (Hacker et al., 2023). Bommasani et al. assessed whether foundation model providers, such as OpenAI, Google and Meta, comply with the AI Act's regulations (Bommasani et al., 2023). Furthermore, Küspert et al. emphasised the need for future-proof governance to address the challenges posed by rapidly evolving business models (Küspert et al., 2023). Powers et al. argued that regulations alone are insufficient and must be considered within a broader context (Powers et al., 2023).

Despite these contributions, the regulatory landscape for foundation models remains underdeveloped, requiring further discussion to enhance AI policy. Moreover, as an important regulation of high-risk AI systems, ex ante conformity assessment is an alternative approach to governing foundation models, making it necessary to analyse and forecast its effects carefully.

2.3. Innovation ecosystems

Innovation ecosystems are defined as 'the collaborative arrangements through which firms combine their individual offerings into a coherent, customer-facing solution' (Adner, 2006), and its concept centres on 'a focal innovation and the set of components (upstream) and complements (downstream) that support it' (Jacobides et al., 2018). Typically, the key elements of an innovation ecosystem revolve around a specific innovation or new value proposition and the constellation of actors supporting it (Jacobides et al., 2018). This holistic value embedded in products and services for end users integrates the innovation efforts of various actors (Dedehayir et al., 2018; Gomes et al., 2018). Furthermore, the ecosystem leader, holding the focal innovation, plays a fundamental role in value co-creation and the development and evolution of the innovation ecosystem (Dedehayir et al., 2018; Gawer & Cusumano, 2014).

The concept of innovation ecosystems not only provides a theoretical foundation to think about the relationship between the core innovation and other actors, but is also widely applied in the field of AI innovation and development and relevant policies (Hannigan et al., 2022). Arenal et al. summarised how the AI innovation ecosystem in China is being developed by identifying key aspects related to the flows of skills, knowledge and funding and the interactions among them (Arenal et al., 2020). Heng et al. discerned the practical effects of AI in Global South countries by depicting the AI ecosystems and offered recommendations to promote AI innovation (Heng et al., 2022). Sun et al. constructed an evaluation system to measure and analyse the spatiotemporal distribution and dynamic evolution of AI development based on innovation ecosystems theory (Sun et al., 2024). Moreover, Powers et al. explored the relationship between AI regulation, trust and advanced AI systems from an ecosystem perspective (Powers et al., 2023).

Although these studies offer valuable insights into AI's future development through the lens of innovation ecosystems theory, their discussion of trustworthiness remains limited. Since trustworthiness is a crucial goal of AI technology and a key driver of AI innovation, it should be integrated into discussions of the AI innovation ecosystem. This can enrich research on AI innovation ecosystems and enhance trustworthiness through innovation ecosystems. Stahl proposed that it is necessary to apply responsible research and innovation to the innovation ecosystems to shape and realise responsible AI innovation ecosystems (Stahl, 2022, 2023).

Compared with other AI systems, the trustworthiness of foundation models is crucial due to their profound impacts on entire ecosystems. The reason is that the unique features, homogenisation and emergence, make foundation models have greater influence and higher trustworthiness risks. Homogenisation means that any flaws in the foundation model are inherited blindly by all adapted models (Bommasani et al.,

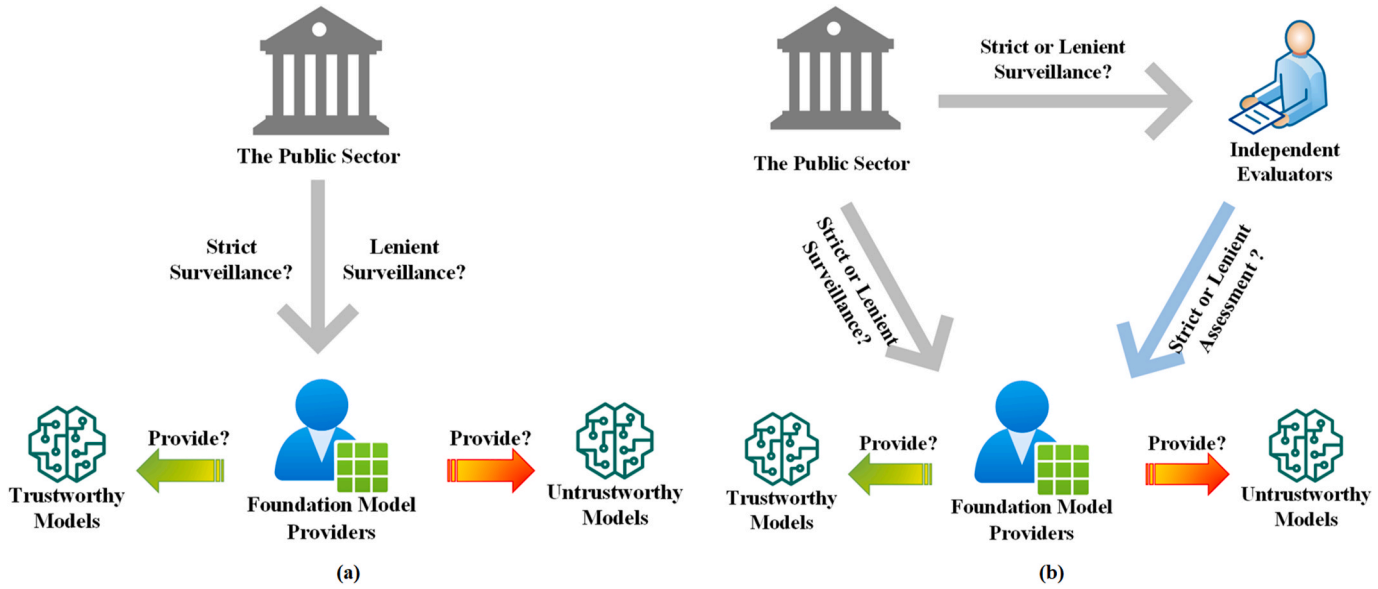


Fig. 1. Two hypothetical scenarios of self-assessment (a) and independent conformity assessment (b).

2022). Emergence indicates that unclear construction and large parameter magnitude make the model difficult to understand and it has unexpected failure modes (Bommasani et al., 2022). The protean nature and unmapped capabilities of foundation models make their innovation ecosystem impacts complex, requiring particular attention (Bommasani et al., 2022; Madiega, 2023).

2.4. Evolutionary game theory

'Evolutionary game theory is about the emergence, transformation, diffusion and stabilization of forms of behavior' (Gintis, 2009). Integrating Darwinian concepts with traditional game theory, this approach was initially employed to elucidate group behaviour in animals, emphasising the crucial role of time evolution (Gintis, 2009; Tanimoto, 2015). A key distinction from traditional game theory is its assumption of bounded rationality, as opposed to the perfect rationality assumption in traditional game theory (Weibull, 1995). Notably, bounded rationality arises primarily from information distribution rather than the degree of rationality (Gintis, 2009).

An evolutionary game comprises multiple-stage games, where player interactions differ from the one-shot or repeated games of classical game theory (Gintis, 2009). Instead, agents are randomly and repeatedly paired, selecting strategies based on contextual suitability rather than past interactions (Gintis, 2009). Meanwhile, in multi-population settings, an agent from each population is randomly selected and paired to participate in each stage game (Weibull, 1995).

Agents adapt and refine their strategies through trial and error, replicating those of more successful counterparts whose strategies yield above-average payoffs within the given context (Gintis, 2009; Tanimoto, 2015). These evolutionary dynamics are captured by replicator equations, where strategies function as replicators. This iterative process leads to the gradual diffusion of the winning strategy across a population, thereby resulting in the establishment of an evolutionary stable strategy (ESS) (Gintis, 2009; Weibull, 1995). Once an ESS is adopted by an entire population, it becomes resistant to invasion by small groups introducing a mutant strategy (Gintis, 2009).

The evolutionary game approach is widely used in regulatory research including AI regulations. For example, Alalawi et al. constructed a tripartite evolutionary game model to analyse the strategy choices of users, AI creators and regulators to identify the possible effects of different regulatory regimes (Alalawi et al., 2024). Bova et al. used an evolutionary game theory model to depict the interactions among governments, auditors, and AI companies, and to explore how governments can deter reckless behaviour through regulatory markets (Bova et al., 2023). Han et al. combined a baseline game-theoretical model, referred to as the DSAIR model, with an evolutionary game approach to demonstrate the potential of a regulatory approach with voluntary commitment (Han et al., 2022). Han et al. also used the DSAIR model and evolutionary game model to analyse the effects of punishment and reward in innovation races (Han et al., 2021). These studies are instructive for the analysis of AI regulation by the evolutionary game method.

In this study, we employ an evolutionary game approach to analyse and forecast the behaviours of key actors within innovation ecosystems, including foundation model providers, the public sector and independent evaluators. The major difference in this study is that we analyse the regulation effects at the level of innovation ecosystems and consider both regulatory relationships and ecological relationships. The analyses are conducted within two hypothetical scenarios of ex ante conformity assessment regulations. The aim is to assess the impact of self-assessment and independent assessment, along with related market surveillance, and to pinpoint the crucial factors influencing the development of trustworthy foundation models. There are three reasons for adopting this approach. First, it is widely applied to analysing regulation issues, aligning well with our research questions. Second, all participants exhibit bounded rationality, seeking optimal strategies within given conditions through continuous adjustment and imitation. Third, the system results evolve through players' constant learning and attempts over time, which can provide insights for the formulation of future-proof policies.

3. Model construction

3.1. Scenario description

In this section, we construct two hypothetical scenarios of self-assessment and independent conformity assessment and analyse the effects of these two regulation methods on the achievement of trustworthy foundation models. According to the AI Act, self-assessment is the major measure of ex ante conformity assessment of high-risk AI systems. Furthermore, independent conformity assessments for biometric identification should be carried out by a notified body (European Commission, 2024), in line with the requirements for CE marking (European Union, 2023). In Scenario 1, foundation model providers

Table 1
The meanings of parameters in Scenario 1.

Scenario 1		
Game agents	Parameters	Meanings
Providers	*MR*	Incomes of foundation models
	TMC	Direct costs of providing trustworthy offerings
	UMC	Direct costs of providing untrustworthy offerings
	I	Indemnities
Public sector	*PB*	Public benefits brought by trustworthy models
	PC	Public costs brought by untrustworthy models
	SSC	Direct costs of strict market surveillance
	EII	Investment in evaluation innovation
	LSC	Direct costs of lenient market surveillance
	r_m	Rate of administrative fines for untrustworthy models
Ecosystem level	*EIC*	Overall costs of evaluation innovation
	R	Ecosystem benefits brought by trustworthy models
	NE	Negative ecosystem effects caused by untrustworthy models

conduct a self-assessment of trustworthiness under the surveillance of the public sector before model launch (Fig. 1 (a)). In Scenario 2, the trustworthiness of the models is assessed by independent evaluators, and both the providers and the evaluators are under the surveillance of the public sector (Fig. 1 (b)). Meanwhile, an internal audit would be conducted. Moreover, the trustworthiness of foundation models can significantly influence their ecosystems; hence, ecosystem effects are considered in these two scenarios.

We apply evolutionary game theory to predict actors' possible behaviour in these two scenarios to extrapolate the regulation effect. The reasons for this are as follows. First, the major actors in these two regulation scenarios would strategically interact with each other attempting to maximise their payoffs. Therefore, game theory provides an appropriate perspective to understand their behaviour in the given situations. Second, since foundation models develop at exponential speed, it would be better to discuss future-proofed regulations (Küspert et al., 2023). Furthermore, being different from traditional game theory, evolutionary game theory merges the idea of time evolution (Tanimoto, 2015); hence, it is more suitable to analyse the potential effects of these regulations. Third, in evolutionary game theory, players are regarded as bounded rational decision-makers instead of perfectly rational ones in traditional games (Gintis, 2009; Weibull, 1995), and this fundamental assumption is closer to reality.

3.2. Model assumption

3.2.1. Scenario 1 for providers' self-assessment

The first scenario involves using self-assessment as an ex ante conformity assessment method to encourage foundation model providers to deliver trustworthy offerings. To analyse the potential effects of this regulation, we establish the following assumptions based on an evolutionary game approach.

Assumption 1. Two main populations—foundation model providers and the public sector—engage in a two-player evolutionary game comprising numerous stage games. In each period, an agent from one population is randomly paired with an agent from the other population to play a stage game using a pure strategy (Gintis, 2009; Weibull, 1995). Over time, agents replicate strategies that yield above-average payoffs within the given context, eventually stabilising at optimal strategies (Gintis, 2009).

Assumption 2. Foundation model providers and the public sector would have two strategies. The providers could adopt the strategy of offering trustworthy foundation models, a choice referred to as PTM, or alternatively, pursue untrustworthy offerings, referred to as PUM. If providers choose the PTM strategy, they can dedicate efforts to developing, operating and innovating trustworthy foundation models. This involves conducting comprehensive self-assessment and innovating evaluation methods and audit tools. Conversely, selecting the PUM strategy would indicate neglecting the models' trustworthiness and ignoring evaluation innovation. Providers opting for this strategy might use falsified documents to attract consumers; however, these untrustworthy offerings would result in losses for clients. Consequently, they would need to compensate their downstream providers and may face administrative fines.

The public sector could conduct strict market surveillance, a strategy referred to as CSS, or adopt a more lenient market surveillance, referred to as CLS. Adopting the CSS strategy entails vigilant efforts to prevent and investigate non-compliant models. The public sector would impose penalties on providers pursuing the PUM strategy, while simultaneously investing in evaluation innovation. Opting for the CLS strategy signifies a minimal regulatory approach to foundation model trustworthiness. Since the regulation of advanced AI technologies is extremely costly (Alalawi et al., 2024), the public sector has an incentive to do so. Consequently, it might fail to detect untrustworthy offerings entering the market and make no investment in evaluation innovation. Therefore, providers opting for the PUM strategy may not face fines in this case, despite offering untrustworthy models.

Assumption 3. The proportion of the providers choosing the PTM strategy would be x_1 $(0 \leq x_1 \leq 1)$, with the remainder selecting the PUM strategy being $1 - x_1$. Regarding the public sector, the proportion adopting CSS strategy would be y_1 $(0 \leq y_1 \leq 1)$, and the CLS strategy would be used by the remaining segment $1 - y_1$. Since players replicate strategies that yield above-average payoffs, the proportion of a population adopting a particular strategy evolves until it reaches a stable equilibrium state (Tanimoto, 2015). Hence, x_1 and y_1 are the functions of time, signifying that they change over time.

Assumption 4. The providers would receive incomes, denoted as *MR*, from providing foundation model offerings to their downstream providers and end users. When the providers employ the PTM strategy, direct costs, denoted as *TMC*, would include payments for the development and operation of trustworthy foundation models and self-assessment. Self-assessment serves not only for conformity assessment but also for internal control of trustworthiness. The direct costs of providing untrustworthy offerings, represented by *UMC* ($UMC < TMC$), would consist of payments for the development and operation of untrustworthy services and falsifying and forging evaluation documents. When untrustworthy services enter the market, the indemnities that the providers must give to compensate the losses of their consumers would be *I*.

Assumption 5. The public sector would benefit from the supply of trustworthy foundation models, as those offerings could promote the development of the digital economy and society on a broad scale. This, in turn, would increase tax revenue and social benefits, with all of these benefits denoted as *PB*. However, once untrustworthy offerings access the market, regardless of the strategy that the public sector adopts, it must take measures to clean up the industry and deal with related social issues. The costs of handling all negative public effects borne by it are denoted as *PC*. When the public sector adopts the CSS strategy, it allocates sufficient resources to market investigation, and this expenditure is represented by *SSC*. Simultaneously, the public sector would invest in evaluation innovation to improve audit methods and tools for all actors in the entire ecosystem, with this investment denoted as *EII*. In this case, they would further fine the providers for their non-compliance behaviour with the rate of r_m. The choice of CLS strategy indicates that they would neither gain sufficient information of trustworthiness nor punish the providers using the PUM strategy due to a lack of resources invested in market investigation. The costs of this regulatory mode are denoted as *LSC* ($LSC < SSC$).

Assumption 6. Achieving trustworthy foundation models requires the

Table 2
Meanings of parameters in Scenario 2.

Scenario 2		
Game agents	Parameters	Meanings
Providers	MR	Incomes of foundation models (same as that in Scenario 1)
	TMC	Direct costs of providing trustworthy offerings (same as that in Scenario 1)
	UMC	Direct costs of providing untrustworthy offerings (same as that in Scenario 1)
	I	Indemnities (same as that in Scenario 1)
	TC_1	Time costs of a launch delay
Public sector	PB	Public benefits brought by trustworthy models (same as that in Scenario 1)
	PC	Public costs incurred by untrustworthy models (same as that in Scenario 1)
	SSC	Direct costs of strict surveillance (same as that in Scenario 1)
	EII	Investment in evaluation innovation (same as that in Scenario 1)
	LSC	Direct costs of lenient surveillance (same as that in Scenario 1)
	r_m	Rate of administrative fines for untrustworthy foundation models (same as that in Scenario 1)
	r_a	Rate of administrative fines for lenient assessment
	TC_2	Time costs of a launch delay
Independent evaluators	AR	Incomes of independent assessment service
	SAC	Direct costs of strict assessment
	LAC	Direct costs of lenient assessment
Ecosystem level	EIC	Overall costs of evaluation innovation (same as that in Scenario 1)
	R	Ecosystem benefits brought by trustworthy models (same as that in Scenario 1)
	NE	Negative ecosystem effects caused by untrustworthy models (same as that in Scenario 1)
	λ	Shared coefficient of evaluation innovation costs between foundation model providers and independent evaluators

development of matching audit methods and tools, necessitating the collective efforts of all stakeholders in the ecosystems. The overall costs of evaluation innovation, denoted as *EIC*, could be covered by the providers when they adopt the PTM strategy and the public sector uses the CLS strategy. However, if the public sector chooses the CSS strategy, the providers would ensure trustworthy services by paying only $EIC - EII$ for evaluation innovation.

Assumption 7. Trustworthy foundation models will attract increased downstream providers to deploy them into various vertical industries. This can promote the rapid evolution of ecosystems and increase their overall welfare, allowing all actors to enjoy ecosystem benefits denoted as *R*. However, untrustworthy offerings could produce significant adverse impacts on numerous stakeholders and industries due to their underlying nature. This could result in economic losses and hinder the adoption of the models, causing the ecosystems to stagnate and even shrink. Thus, untrustworthy services could have negative effects on all actors in ecosystems denoted as *NE*.

Table 1 lists all the parameter definitions used in Scenario 1.

3.2.2. Scenario 2 for independent conformity assessment

In Scenario 2, the ex ante conformity assessment of foundation models would be conducted by independent evaluators. We propose the following assumptions and explain the similarities and differences between scenarios 1 and 2.

Assumption 8. In addition to foundation model providers and the public sector, another main player in this scenario would be independent evaluators. Similarly, the three groups would also interact with each other strategically and constantly optimise their strategy choices until stabilising at the best ones in the given context.

Assumption 9. In this scenario, foundation model providers and the public sector have the same strategic choices—(PTM, PUM) and (CSS, CLS)—as in Scenario 1. When foundation model providers adopt the PTM strategy, they would also conduct an internal evaluation to ensure trustworthiness and innovate actively on audit methods and tools. Independent evaluators could adopt one of two assessment strategies: a strict assessment, referred to as MSA, or a lenient assessment strategy, referred to as MLA. Under the MSA strategy, they would identify untrustworthy services effectively and prevent these models from launching. Conversely, if they adopt the MLA strategy, they would fail to recognise untrustworthy models, allowing them to enter the market. Additionally, evaluators following the MLA strategy might engage in document forgery. It is important to note that independent evaluators must also be regulated by the public sector. If untrustworthy models are placed on the market, the public sector would impose fines on both providers and independent evaluators. However, if providers supply trustworthy services, evaluators would not be penalised, as lenient assessments are difficult to detect.

Assumption 10. The proportion of the providers using the PTM strategy would be x_2 $(0 \leq x_2 \leq 1)$ and the rest adopting the PUM strategy would be $1 - x_2$. The proportion of the public sector with CSS would be y_2 $(0 \leq y_2 \leq 1)$, while the CLS strategy would be used by the remaining segment $1 - y_2$. Furthermore, the proportion of independent evaluators choosing the MSA strategy would be z_2 $(0 \leq z_2 \leq 1)$, while the remainder adopting the MLA strategy would be $1 - z_2$. Similar to x_1 and y_1 in Scenario 1, x_2, y_2 and z_2 are the functions of time and change with time.

Assumption 11. Similar to Scenario 1, under the requirement of independent assessment, the providers would receive income *MR* for supplying foundation model services. They would afford the costs *TMC* when choosing the PTM strategy, cover the costs *UMC* ($UMC < TMC$) with the PUM strategy and provide the indemnities *I* to their clients for untrustworthy models. The major difference between these two regulatory scenarios is that the providers would have to pay independent evaluators for their services. Moreover, since third-party audit is still under development at present, it would take time for the regulation of independent assessment to be implemented. Therefore, the providers would bear time costs TC_1 while waiting for external evaluation.

Assumption 12. In this scenario, the public sector would also enjoy the benefits *PB* of trustworthy foundation models, pay the costs *PC* for untrustworthy offerings, afford the expenditure *SSC* with strict market surveillance, make an investment *EII* in evaluation innovation and cover the costs *LSC* with lenient surveillance. When untrustworthy foundation models enter the market, they would fine the providers at the rate of r_m and independent evaluators at the rate of r_a. Moreover, the public sector would also bear time costs TC_2 due to the delay in launching foundation models.

Assumption 13. The evaluators would have incomes *AR* ($AR < MR$) for providing independent assessment services. The costs of strict assessment would be *SAC*, and the costs of lenient assessment including forging documents would be *LAC* ($LAC < SAC$).

Assumption 14. As with Scenario 1, all actors would enjoy the ecosystem benefits *R* brought by trustworthy foundation models and be affected by the negative ecosystem impacts *NE* of untrustworthy offerings. There would also be the overall costs of evaluation innovation *EIC* at the ecosystem level, but the providers and the evaluators would share the costs of the remainder after the public sector's investment with coefficient λ.

All parameter definitions for Scenario 2 are provided in Table 2.

Table 3
Payoff matrix in Scenario 1.

Strategies		Payoffs accruing to player	
Providers	Public sector	Providers	Public sector
PTM (x_1)	**CSS** (y_1)	$R+MR-TMC-(EIC-EII)$	$R+PB-SSC-EII$
PTM (x_1)	**CLS** ($1-y_1$)	$R+MR-TMC-EIC$	$R+PB-LSC$
PUM ($1-x_1$)	**CSS** (y_1)	$-NE+MR-UMC-I-r_mMR$	$-NE+r_mMR-PC-SSC-EII$
PUM ($1-x_1$)	**CLS** ($1-y_1$)	$-NE+MR-UMC-I$	$-NE-PC-LSC$

Table 4
Payoff matrix in Scenario 2.

Strategies			Payoffs accruing to player		
Providers	Public sector	Independent evaluators	Providers	Public sector	Independent evaluators
PTM (x_2)	**CSS** (y_2)	**MSA** (z_2)	$R+MR-TC_1-AR-TMC-\lambda(EIC-EII)$	$R+PB-TC_2-SSC-EII$	$R+AR-SAC-(1-\lambda)(EIC-EII)$
PTM (x_2)	**CSS** (y_2)	**MLA** ($1-z_2$)	$R+MR-TC_1-AR-TMC-(EIC-EII)$	$R+PB-TC_2-SSC-EII$	$R+AR-LAC$
PTM (x_2)	**CLS** ($1-y_2$)	**MSA** (z_2)	$R+MR-TC_1-AR-TMC-\lambda EIC$	$R+PB-TC_2-LSC$	$R+AR-SAC-(1-\lambda)EIC$
PTM (x_2)	**CLS** ($1-y_2$)	**MLA** ($1-z_2$)	$R+MR-TC_1-AR-TMC-EIC$	$R+PB-TC_2-LSC$	$R+AR-LAC$
PUM ($1-x_2$)	**CSS** (y_2)	**MSA** (z_2)	$-AR-UMC$	$-SSC-EII$	$AR-SAC-(EIC-EII)$
PUM ($1-x_2$)	**CSS** (y_2)	**MLA** ($1-z_2$)	$-NE+MR-TC_1-AR-UMC-I-r_mMR$	$-NE+r_mMR+r_aAR-TC_2-PC-SSC-EII$	$-NE+AR-LAC-r_aAR$
PUM ($1-x_2$)	**CLS** ($1-y_2$)	**MSA** (z_2)	$-AR-UMC$	$-LSC$	$AR-SAC-EIC$
PUM ($1-x_2$)	**CLS** ($1-y_2$)	**MLA** ($1-z_2$)	$-NE+MR-TC_1-AR-UMC-I$	$-NE-TC_2-PC-LSC$	$-NE+AR-LAC$

3.3. Payoff matrix and replicator dynamics equation

3.3.1. The payoff matrix and replicator dynamics equation in scenario 1

Based on Assumptions 1–7 in the regulation scenario of self-assessment, there would be four types of games between foundation model providers and the public sector, and the payoff matrix of two-player populations is shown in Table 3.

According to Table 3, the expected payoffs for foundation model providers choosing the PTM strategy and those adopting the PUM strategy, as well as their overall average profit in this scenario would be as follows, respectively.

$$U_{p11}=y_1[R+MR-TMC-(EIC-EII)]+(1-y_1)(R+MR-TMC-EIC) \tag{1}$$

$$U_{p12}=y_1(-NE+MR-UMC-I-r_mMR)+(1-y_1)(-NE+MR-UMC-I) \tag{2}$$

$$U_{p1}=x_1U_{p11}+(1-x_1)U_{p12} \tag{3}$$

The expected payoffs for the public sector adopting the CSS strategy and the CLS strategy, as well as their overall average profit in this scenario would be as follows, respectively.

$$U_{s11}=x_1(R+PB-SSC-EII)+(1-x_1)(-NE+r_mMR-PC-SSC-EII) \tag{4}$$

$$U_{s12}=x_1(R+PB-LSC)+(1-x_1)(-NE-PC-LSC) \tag{5}$$

$$U_{s1}=y_1U_{s11}+(1-y_1)U_{s12} \tag{6}$$

According to evolutionary game theory, individuals replicate strategies that yield above-average payoffs, and the replicator dynamics function models the evolution of a strategy within a population (Smith, 2012). Thus, the replicator dynamic equation of foundation model providers choosing the PTM strategy and that of the public sector using the CSS strategy would be as follows, respectively.

$$F_1(x_1)=\frac{dx_1}{dt}=x_1\left(U_{p11}-U_{p1}\right)=-x_1(x_1-1)(I-TMC-EIC+UMC+NE+R+y_1EII+y_1r_mMR) \tag{7}$$

$$F_1(y_1)=\frac{dy_1}{dt}=y_1(U_{s11}-U_{s1})=y_1(y_1-1)(EII-LSC+SSC-r_mMR+x_1r_mMR) \tag{8}$$

3.3.2. The payoff matrix and replicator dynamics equation in scenario 2

According to Assumptions 8–14 in the regulation scenario of independent assessment, there would be eight types of games among foundation model providers, independent evaluators and the public sector, and the payoff matrix of the three groups is shown in Table 4.

According to Table 4, the expected payoffs associated with the PTM and PUM strategies, as well as the overall average profit considering both strategic choices would be as follows, respectively.

$$\begin{aligned}U_{p21}=&y_2z_2[R+MR-TC_1-AR-TMC-\lambda(EIC-EII)]\\&+y_2(1-z_2)[R+MR-TC_1-AR-TMC-(EIC-EII)]\\&+(1-y_2)z_2(R+MR-TC_1-AR-TMC-\lambda EIC)\\&+(1-y_2)(1-z_2)(R+MR-TC_1-AR-TMC-EIC)\end{aligned} \tag{9}$$

$$\begin{aligned}U_{p22}=&y_2z_2(-AR-UMC)+y_2(1-z_2)(-NE+MR-TC_1-AR-UMC\\&-I-r_mMR)+(1-y_2)z_2(-AR-UMC)\\&+(1-y_2)(1-z_2)(-NE+MR-TC_1-AR-UMC-I)\end{aligned} \tag{10}$$

$$U_{p2}=x_2U_{p21}+(1-x_2)U_{p22} \tag{11}$$

In Scenario 2, the expected payoffs for the public sector employing the CSS strategy and selecting the CLS strategy, as well as the overall average profit across both choices would be as follows, respectively.

Table 5
Values of the Jacobian Matrix (J) elements in Scenario 1.

Equilibrium points	A_{11}	A_{12}	A_{21}	A_{22}
$E_1(0,0)$	$I-TMC-EIC+UMC+NE+R$	0	0	$LSC-EII-SSC+r_mMR$
$E_2(0,1)$	$EII-EIC-TMC+I+UMC+NE+R+r_mMR$	0	0	$EII-LSC+SSC-r_mMR$
$E_3(1,0)$	$EIC+TMC-I-UMC-NE-R$	0	0	$LSC-EII-SSC$
$E_4(1,1)$	$EIC-EII+TMC-I-UMC-NE-R-r_mMR$	0	0	$EII-LSC+SSC$

Note: $E_4(1,1)$ cannot be an ESS because its $A_{22}>0$ making it impossible to satisfy both conditions simultaneously.

Table 6
Evolutionary stable strategy and corresponding conditions in Scenario 1.

Case	Condition	ESS
Case 1	$EIC+TMC-R>UMC+I+NE$ $SSC+EII-r_mMR>LSC$	$E_1(0,0)$
Case 2	$(EIC-EII)+TMC-R>UMC+I+r_mMR+NE$ $LSC>SSC+EII-r_mMR$	$E_2(0,1)$
Case 3	$UMC+I+NE>EIC+TMC-R$	$E_3(1,0)$

$$U_{s21}=x_2z_2(R+PB-TC_2-SSC-EII)+x_2(1-z_2)(R+PB-TC_2-SSC-EII)+(1-x_2)z_2(-SSC-EII)+(1-x_2)(1-z_2)(-NE+r_mMR+r_aAR-TC_2-PC-SSC-EII) \quad (12)$$

$$U_{s22}=x_2z_2(R+PB-TC_2-LSC)+x_2(1-z_2)(R+PB-TC_2-LSC)+(1-x_2)z_2(-LSC)+(1-x_2)(1-z_2)(-NE-TC_2-PC-LSC) \quad (13)$$

$$U_{s2}=y_2U_{s21}+(1-y_2)U_{s22} \quad (14)$$

As for independent evaluators, the expected payoffs of choosing the MSA and MLA strategies, as well as the overall average profits of using the two strategies would be as follows, respectively.

$$U_{e21}=x_2y_2[R+AR-SAC-(1-\lambda)(EIC-EII)]+x_2(1-y_2)[R+AR-SAC-(1-\lambda)EIC]+(1-x_2)y_2[AR-SAC-(EIC-EII)]+(1-x_2)(1-y_2)(AR-SAC-EIC) \quad (15)$$

$$U_{e22}=x_2y_2(R+AR-LAC)+x_2(1-y_2)(R+AR-LAC)+(1-x_2)y_2(-NE+AR-LAC-r_aAR)+(1-x_2)(1-y_2)(-NE+AR-LAC) \quad (16)$$

Table 7
Eigenvalues of the Jacobian matrix (J) in Scenario 2.

Equilibrium point	Eigenvalue			Symbols of the three eigenvalues
	λ_1	λ_2	λ_3	
$E'_1(0,0,0)$	$LAC-EIC+NE-SAC$	$LSC-EII-SSC+r_mMR+r_aAR$	$I-TMC-EIC+UMC+NE+R$	$(\pm,\pm,\pm)$
$E'_2(0,0,1)$	$LSC-EII-SSC$	$EIC-LAC-NE+SAC$	$UMC-TMC+R+MR-TC_1-\lambda EIC$	$(-,\pm,\pm)$
$E'_3(0,1,0)$	$EII-LSC+SSC-r_mMR-r_aAR$	$EII-EIC+LAC+NE-SAC+r_aAR$	$EII-EIC-TMC+I+UMC+NE+R+r_mMR$	$(\pm,\pm,\pm)$
$E'_4(0,1,1)$	$EII-LSC+SSC$	$EIC-EII-LAC-NE+SAC-r_aAR$	$UMC-TMC+R+MR-TC_1-\lambda EIC+\lambda EII$	$(+,\pm,\pm)$
$E'_5(1,0,0)$	$LSC-EII-SSC$	$LAC-EIC-SAC+\lambda EIC$	$EIC+TMC-I-UMC-NE-R$	$(-,-,\pm)$
$E'_6(1,0,1)$	$LSC-EII-SSC$	$EIC-LAC+SAC-\lambda EIC$	$TMC-UMC-R-MR+TC_1+\lambda EIC$	$(-,+,\pm)$
$E'_7(1,1,0)$	$EII-LSC+SSC$	$EII-EIC+LAC-SAC+\lambda EIC-\lambda EII$	$EIC-EII+TMC-I-UMC-NE-R-r_mMR$	$(+,-,\pm)$
$E'_8(1,1,1)$	$EII-LSC+SSC$	$EIC-EII-LAC+SAC-\lambda EIC+\lambda EII$	$TMC-UMC-R-MR+TC_1+\lambda EIC-\lambda EII$	$(+,+,\pm)$

Table 8
ESS and corresponding conditions in Scenario 2.

Scenario	Condition	ESS
Case 4	$TMC+EIC-R>UMC+I+NE$ $SAC+EIC>LAC+NE$ $SSC+EII-r_mMR-r_aAR>LSC$	$E'_1(0,0,0)$
Case 5	$TMC+\lambda EIC+TC_1-MR-R>UMC$ $LAC+NE>SAC+EIC$	$E'_2(0,0,1)$
Case 6	$TMC+(EIC-EII)-R>UMC+I+r_mMR+NE$ $LSC>SSC+EII-r_mMR-r_aAR$ $SAC+(EIC-EII)>LAC+r_aAR+NE$	$E'_3(0,1,0)$
Case 7	$UMC+I+NE>TMC+EIC-R$	$E'_5(1,0,0)$

Table 9
Initial values of all parameters in the two scenarios.

Scenario 1			Scenario 2		
Game agents	Parameters	Values	Game agents	Parameters	Values
Providers	x_1	0.5	Providers	x_2	0.5
	MR	550		MR	550
	TMC	200		TMC	200
	UMC	140		UMC	140
	I	40		I	40
	r_m	0.02		r_m	0.02
				TC_1	50
			Independent evaluators	y_2	0.5
				AR	180
				SAC	50
				LAC	20
				r_a	0.02
Public sector	z_1	0.5	Public sector	z_2	0.5
	PB	70		PB	70
	SSC	40		SSC	40
	EII	15		EII	15
	LSC	10		LSC	10
	PC	45		PC	45
				TC_2	5
Ecosystem level	EIC	100	Ecosystem level	EIC	100
	R	55		R	55
	NE	55		NE	55
				λ	0.5

$$U_{e2}=z_2U_{e21}+(1-z_2)U_{e22} \quad (17)$$

Similarly, the replicator dynamic equations for the providers adopting the PTM strategy, evaluators using the CSS strategy and the public sector choosing the MSA strategy would be as follows, respectively.

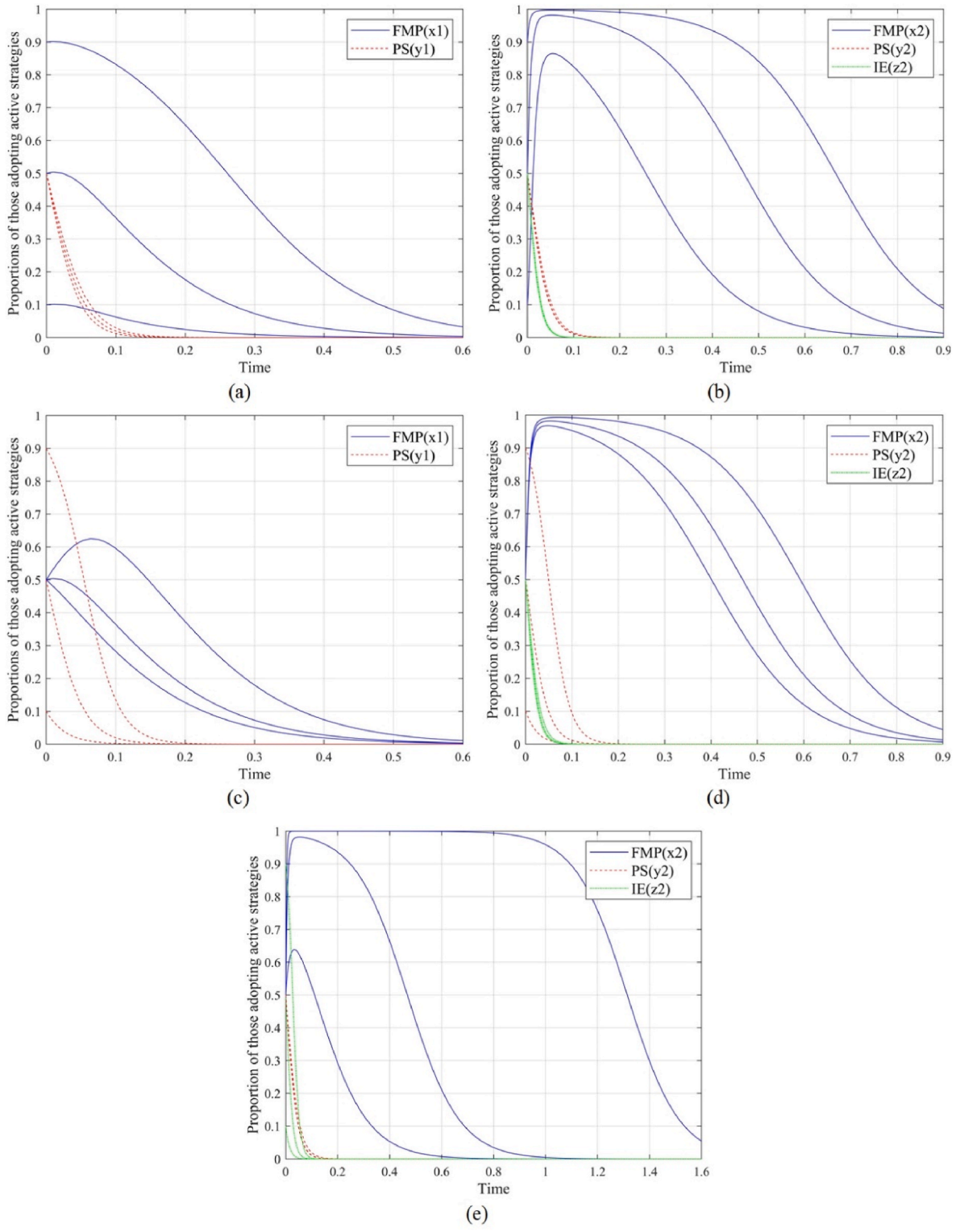


Fig. 2. Effects of the different initial values of x_1 (a), x_2 (b), y_1 (c), y_2 (d) and z_2 (e) on the evolution results. x_1 and x_2 are the proportion of the providers adopting the PTM strategy in Scenarios 1 and 2, respectively; y_1 and y_2 are the proportion of the public sector actors using the CSS strategy in Scenarios 1 and 2, respectively; z_2 is the proportion of independent evaluators choosing the MSA strategy in Scenario 2. Foundation model providers, the public sector and independent evaluators are denoted as FMP, PS and IE, respectively, in the figures.

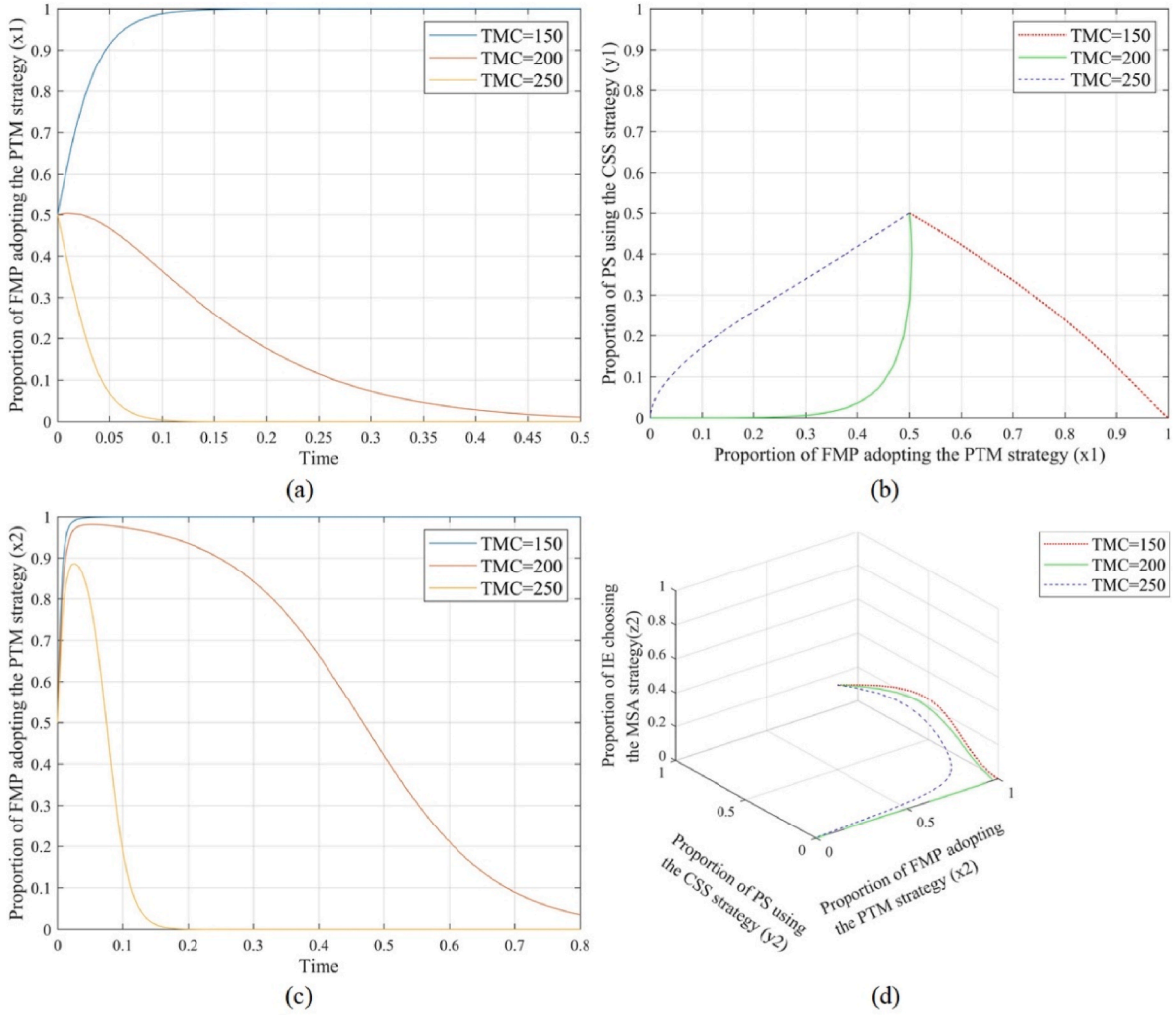


Fig. 3. Effects of *TMC* on the stable strategies of foundation model providers (a) and on the system ESS (b) in Scenarios 1, and the corresponding effects in Scenarios 2 (c–d). *TMC* denotes the direct costs incurred by the providers providing trustworthy models.

$$F_2(x_2) = \frac{dx_2}{dt} = x_2\left(U_{p21} - U_{p2}\right) = -x_2(x_2-1)(I - TMC - EIC + UMC + NE + R + y_2EII + z_2EIC - z_2I - z_2NE + z_2MR - z_2TC_1 - z_2\lambda EIC - y_2z_2EII + y_2r_mMR + y_2z_2\lambda EII - y_2z_2r_mMR) \tag{18}$$

$$F_2(y_2) = \frac{dy_2}{dt} = y_2(U_{s21} - U_{s2}) = y_2(y_2-1)(EII - LSC + SSC - r_mMR - r_aAR + x_2r_mMR + x_2r_aAR + z_2r_mMR + z_2r_aAR - x_2z_2r_mMR - x_2z_2r_aAR) \tag{19}$$

$$F_2(z_2) = \frac{dz_2}{dt} = z_2(U_{e21} - U_{e2}) = z_2(z_2-1)(EIC - LAC - NE + SAC - y_2EII + x_2NE - x_2\lambda EIC - y_2r_aAR + x_2y_2\lambda EII + x_2y_2r_aAR) \tag{20}$$

4. Model analysis

4.1. Stability analysis of equilibrium points in the evolutionary game system

4.1.1. Stability analysis of equilibrium points in the system of scenario 1

Based on the stability theory of ordinary differential equations, we identify four equilibrium points corresponding to pure strategies, $E_1(0, 0)$, $E_2(1, 0)$, $E_3(0, 1)$ and $E_4(1, 1)$, obtained by setting function (7) $F_1(x_1) = \frac{dx_1}{dt}$ and function (8) $F_1(z_1) = \frac{dz_1}{dt}$ to zero. These equilibrium points should be further evaluated for asymptotic stability using the Jacobian Matrix (J) to identify if they are the ESS of the system (Friedman, 1991).

The Jacobian Matrix (J) of the game system in the regulation scenario of self-assessment is as follows:

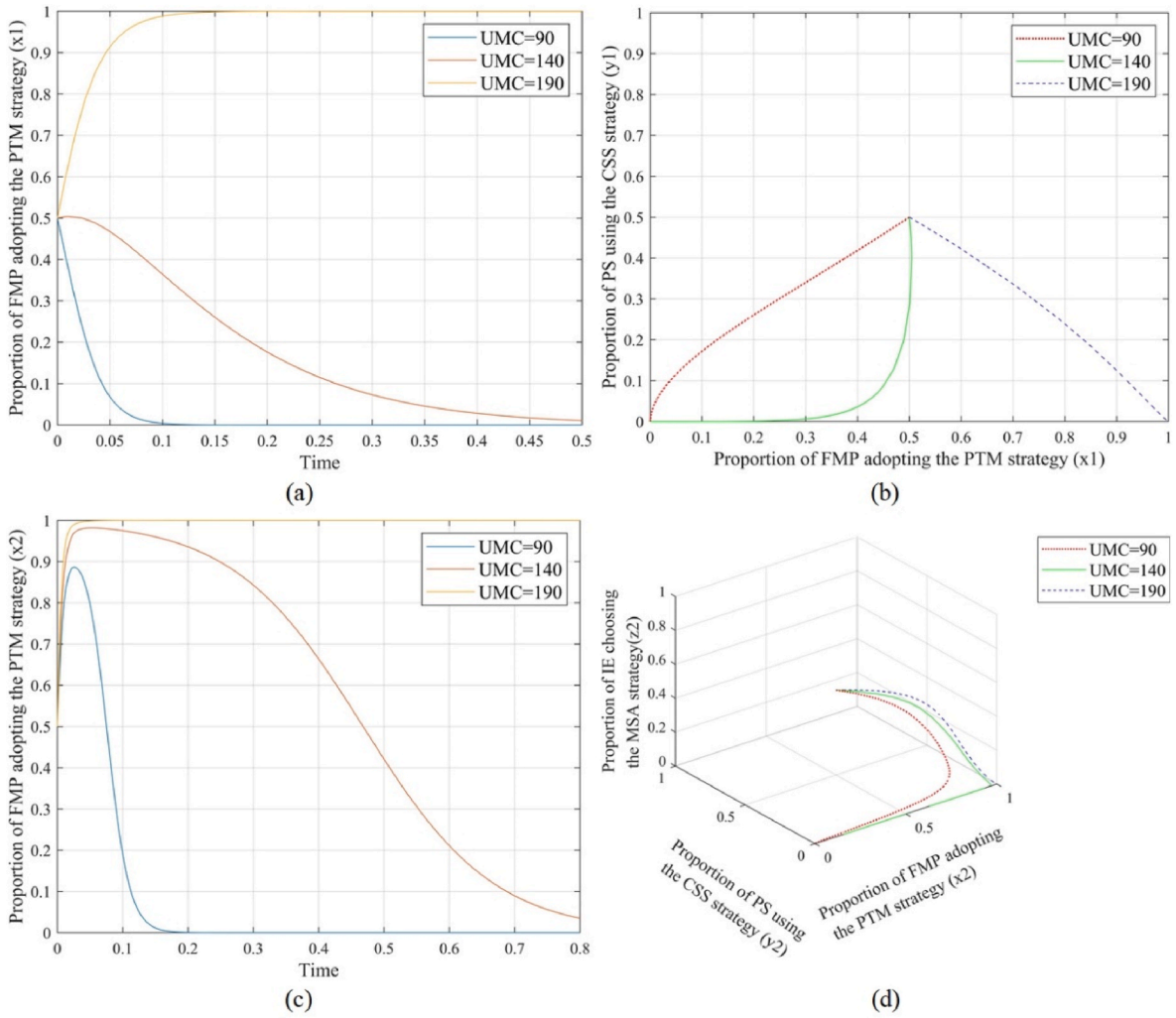


Fig. 4. Effects of *UMC* on the stable strategies of foundation model providers (a) and on the system ESS (b) in scenarios 1, and the corresponding effects in scenarios 2 (c–d). *UMC* denotes the direct costs incurred by the providers providing untrustworthy models.

$$J_1 = \begin{bmatrix} \frac{\partial F_1(x)}{\partial x_1} & \frac{\partial F_1(x)}{\partial z_1} \\ \frac{\partial F_1(z)}{\partial x_1} & \frac{\partial F_1(z)}{\partial z_1} \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \tag{21}$$

where

$$A_{11} = -(2x_1 - 1)(I - TMC - EIC + UMC + NE + R + y_1 EII + y_1 r_m MR) \tag{22}$$

$$A_{12} = -x_1(x_1 - 1)(\mathrm{EII} + r_m \mathrm{MR}) \tag{23}$$

$$A_{21} = y_1(y_1 - 1) r_m \mathrm{MR} \tag{24}$$

$$A_{22} = (2y_1 - 1)(EII - LSC + SSC - r_m MR + x_1 r_m MR) \tag{25}$$

The ESS should meet the requirements of the two conditions, $det(J_1) = A_{11}A_{22} - A_{12}A_{21} > 0$ and $tr(J_1) = A_{11} + A_{22} < 0$. In Table 5, the values of the Jacobian Matrix (J) elements for each equilibrium point are listed. Then, the ESS of this game system and the corresponding conditions are shown in Table 6.

The ideal state is represented as $E_4(1,1)$ which indicates that the providers adopt the PTM strategy; meanwhile, the public sector use the CSS strategy, but it could not be the ESS according to the analysis of asymptotic stability. Thus, a meaningful ESS that could be achieved is $E_3(1,0)$ which represents that the providers offer trustworthy model services while the public sector carries out lenient market surveillance. To reach $E_3(1,0)$, it should meet the condition of Case 3 that the sum of the key costs of providing untrustworthy offerings is larger than the margin of the key costs and benefits when the providers adopt the PTM strategy and the public sector uses the CLS strategy. According to the evolution result, the key costs of choosing the PUM strategy include the direct costs *UMC*, the indemnities given to consumers *I* and the negative ecosystem effects *NE*. In contrast, the key costs and benefits of adopting the PTM strategy comprise the direct costs of providing trustworthy offerings *TMC*, overall costs of evaluation innovation *EIC* and the ecosystem benefits *R*. Furthermore, if the evolution results indicate that providers have adopted the PUM strategy, the surveillance remains completely ineffective regardless of whether the public sector implements the CSS strategy.

4.1.2. Stability analysis of equilibrium points in the system of scenario 2

Similarly, the equilibrium points can be reached when we let function (18) $F_2(x_2) = \frac{dx_2}{dt} = 0$, function (19) $F_2(y_2) = \frac{dy_2}{dt} = 0$ and function (20) $F_2(z_2) = \frac{dz_2}{dt} = 0$, and then, eight equilibrium points of pure strategies, $E_1'(0,0,0)$, $E_2'(0,0,1)$, $E_3'(0,1,0)$, $E_4'(0,1,1)$, $E_5'(1,0,0)$, $E_6'(1,0,1)$, $E_7'(1,1,0)$ and $E_8'(1,1,1)$ are gained.

Likewise, these equilibrium points should also be evaluated for asymptotic stability by using the Jacobian Matrix (J) to identify which are the ESS (Lyapunov, 1992). The Jacobian Matrix (J) of the game

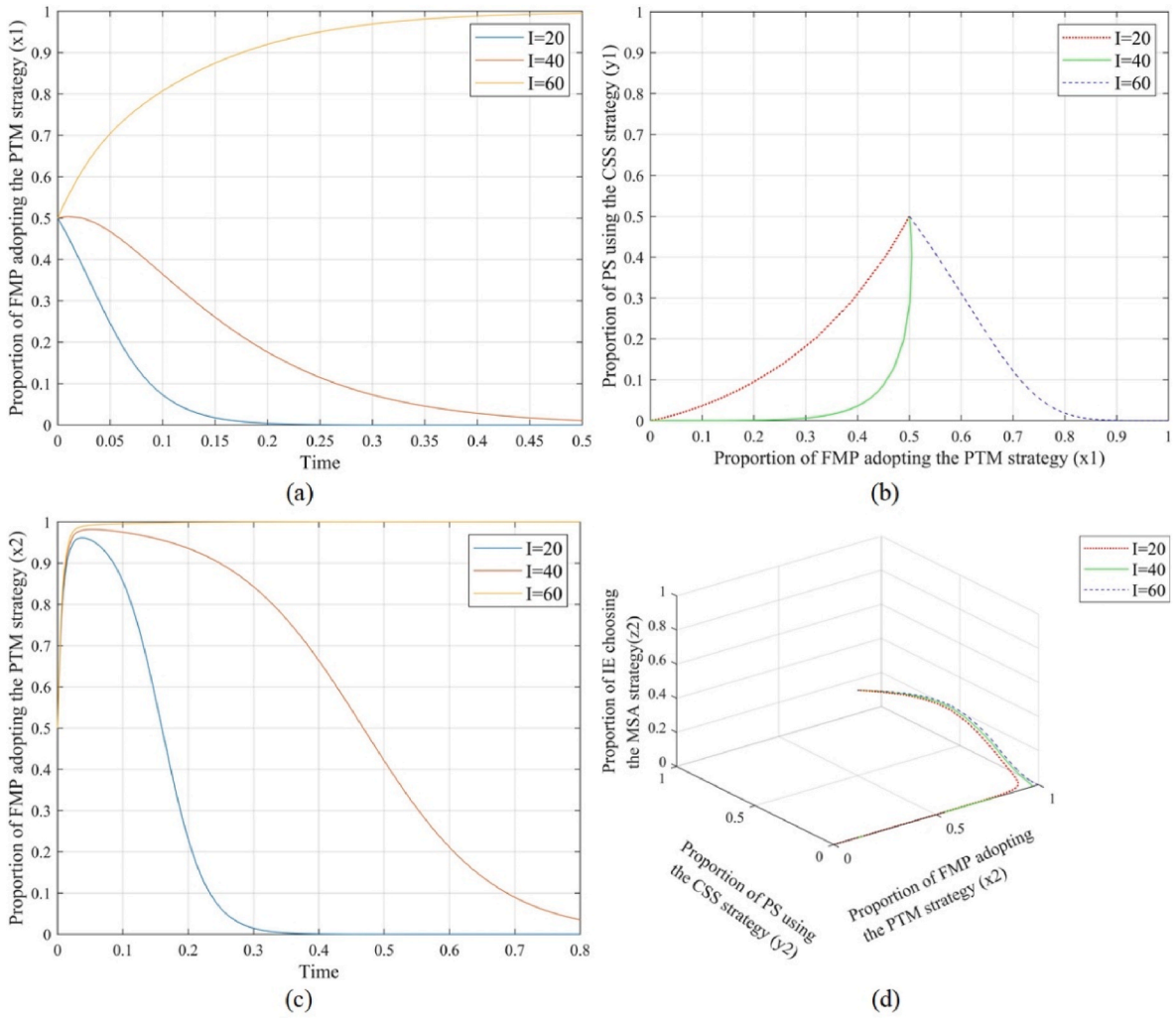


Fig. 5. Effects of I on the stable strategies of foundation model providers (a) and on the system ESS (b) in scenarios 1, and the corresponding effects in scenarios 2 (c–d). I denotes the indemnities given to clients for the provision of untrustworthy offerings.

system of independent assessment is as follows.

$$J_2 = \begin{bmatrix} \frac{\partial F_2(x)}{\partial x_2} & \frac{\partial F_2(x)}{\partial y_2} & \frac{\partial F_2(x)}{\partial z_2} \\ \frac{\partial F_2(y)}{\partial x_2} & \frac{\partial F_2(y)}{\partial y_2} & \frac{\partial F_2(y)}{\partial z_2} \\ \frac{\partial F_2(z)}{\partial x_2} & \frac{\partial F_2(z)}{\partial y_2} & \frac{\partial F_2(z)}{\partial z_2} \end{bmatrix} = \begin{bmatrix} A'_{11} & A'_{12} & A'_{13} \\ A'_{21} & A'_{22} & A'_{23} \\ A'_{31} & A'_{32} & A'_{33} \end{bmatrix} \tag{26}$$

where

$$A'_{11} = -(2x_2-1)\big(I - TMC - EIC + UMC + NE + R + y_2EII + z_2EIC - z_2I - z_2NE + z_2MR - z_2TC_1 - z_2\lambda EIC - y_2z_2EII + y_2r_mMR + y_2z_2\lambda EII - y_2z_2r_mMR\big) \tag{27}$$

$$A'_{12} = -x_2(x_2-1)(EII - z_2EII + r_mMR + z_2\lambda EII - z_2r_mMR) \tag{28}$$

$$A'_{13} = x_2(x_2-1)\big(I - EIC + NE - MR + TC_1 + \lambda EIC + y_2EII - y_2\lambda EII + y_2r_mMR\big) \tag{29}$$

$$A'_{21} = -y_2(y_2-1)(z_2-1)(r_mMR + r_aAR) \tag{30}$$

$$A'_{22} = (2y_2-1)\big(EII - LSC + SSC - r_1MR - r_aAR + x_2r_mMR + x_2r_aAR + z_2r_mMR + z_2r_aAR - x_2z_2r_mMR - x_2z_2r_aAR\big) \tag{31}$$

$$A'_{23} = -y_2(x_2-1)(y_2-1)(r_mMR + r_aAR) \tag{32}$$

$$A'_{31} = z_2(z_2-1)(NE - \lambda EIC + y_2\lambda EII + y_2r_aAR) \tag{33}$$

$$A'_{32} = -z_2(z_2-1)(EII + r_aAR - x_2\lambda EII - x_2r_aAR) \tag{34}$$

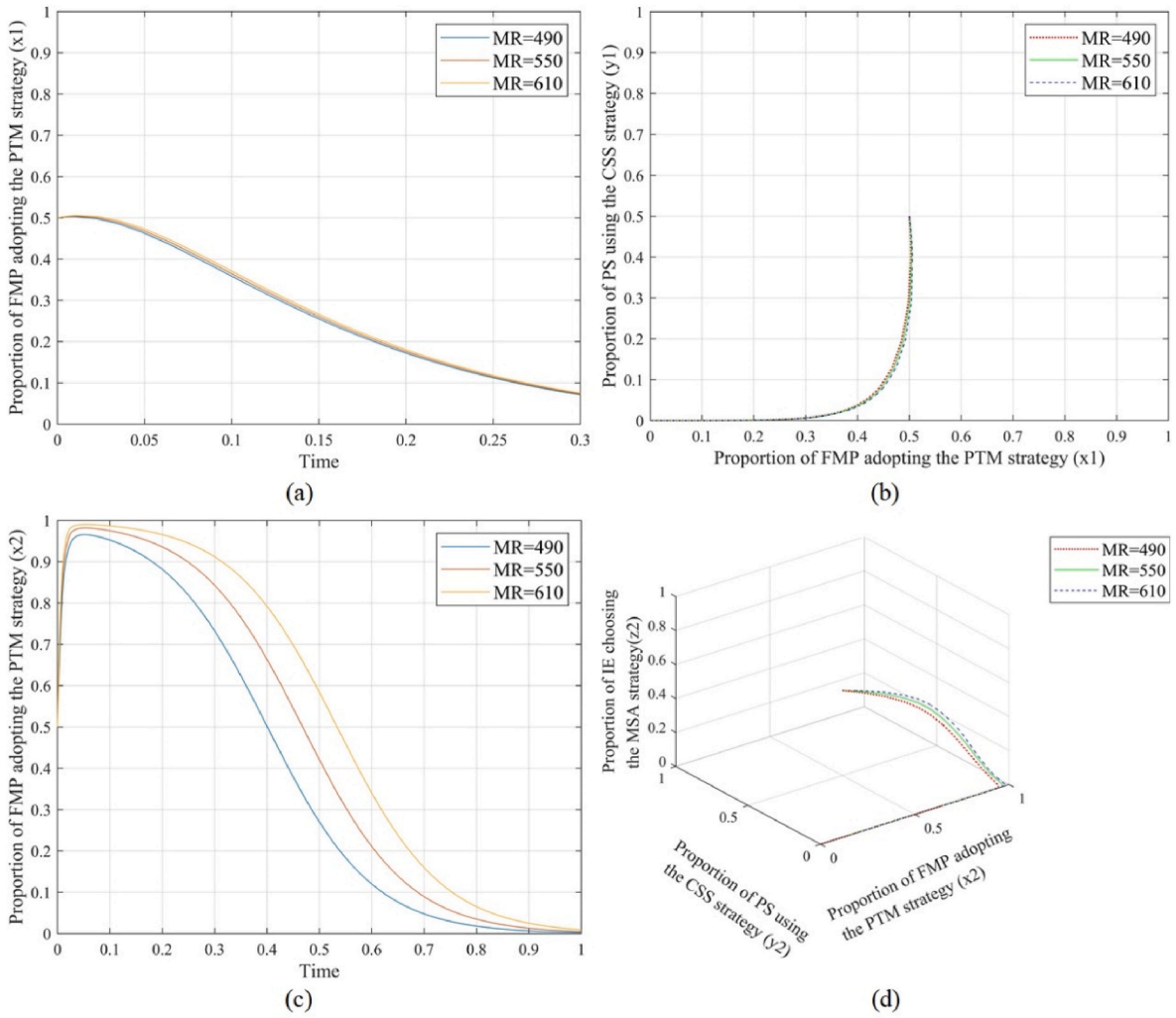


Fig. 6. Effects of *MR* on the stable strategies of foundation model providers (a) and on the system ESS (b) in scenarios 1, and the corresponding effects in scenarios 2 (c–d). *MR* denotes the incomes of foundation models.

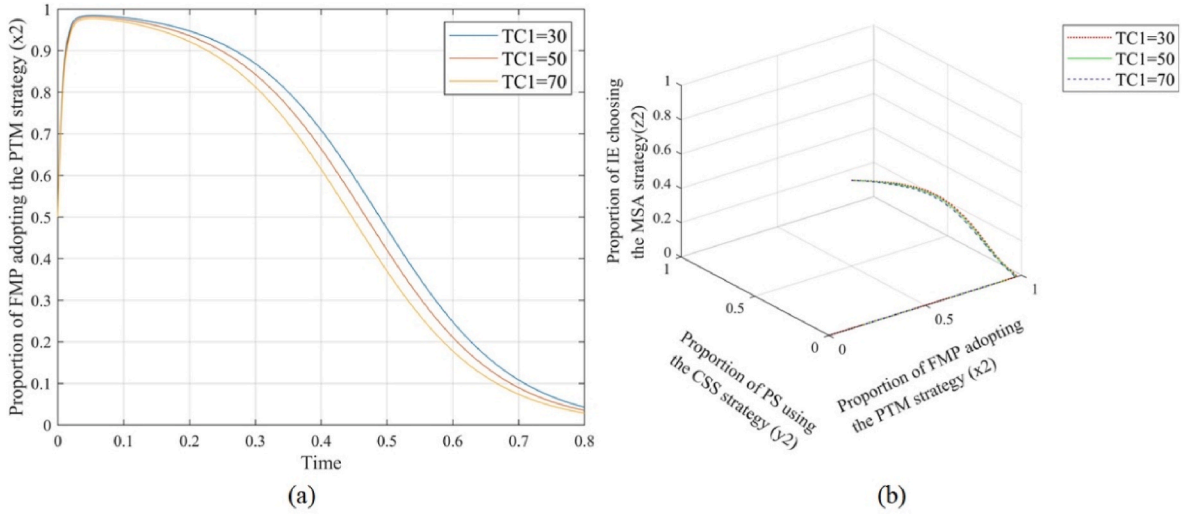


Fig. 7. Effects of TC_1 on the stable strategies of foundation model providers (a) and on the system ESS (b) in Scenario 2. TC_1 denotes the time costs of the launch delay for independent assessment.

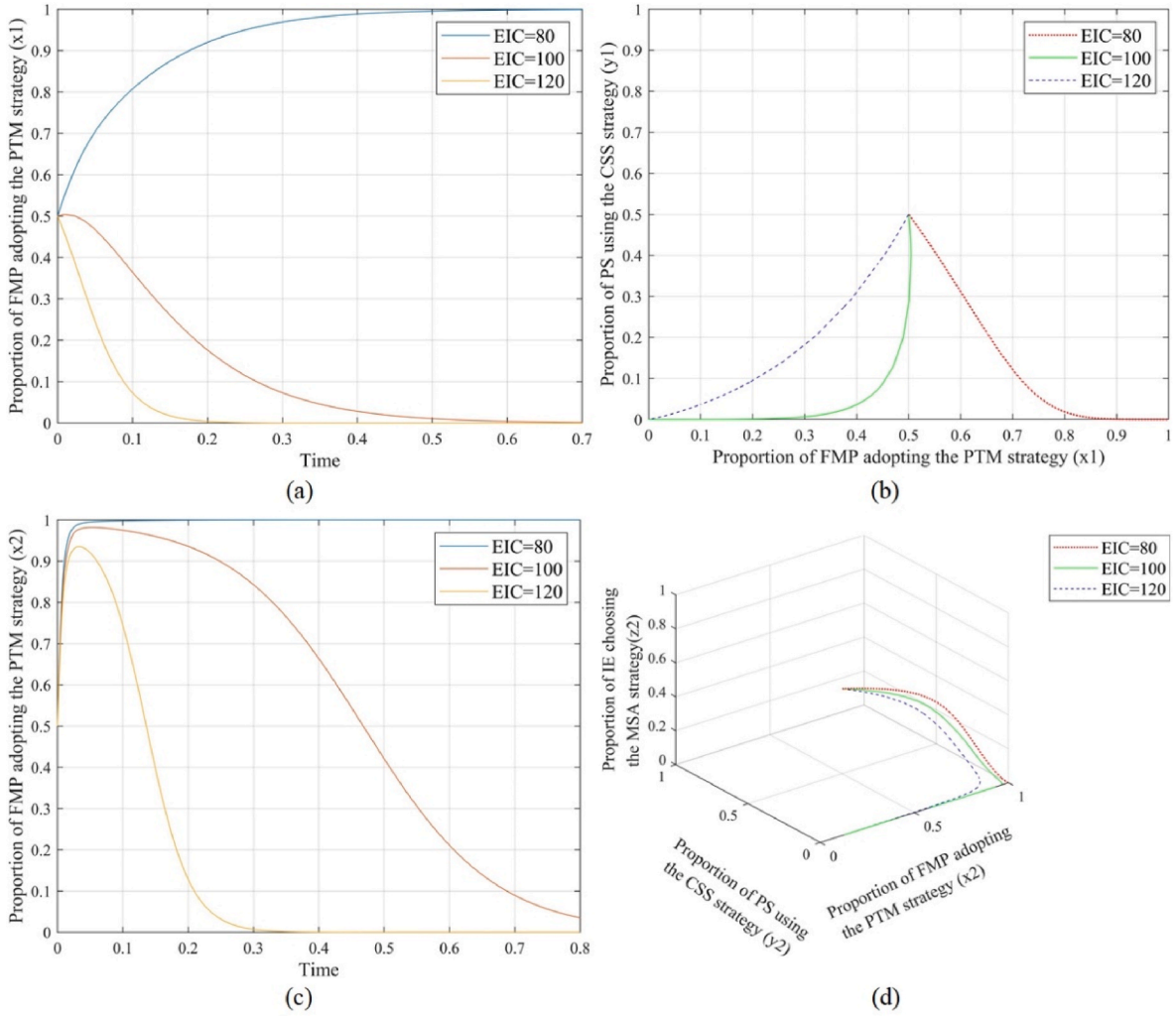


Fig. 8. Effects of *EIC* on the stable strategies of foundation model providers (a) and on the system ESS (b) in scenarios 1, and the corresponding effects in scenarios 2 (c–d). *EIC* denotes the overall costs of evaluation innovation.

$$A_{33}^{'} = (2z_2 - 1)\big(EIC - LAC - NE + SAC - y_2EII + x_2NE - x_2\lambda EIC - y_2r_aAR + x_2y_2\lambda EII + x_2y_2r_aAR\big) \quad (35)$$

According to Lyapunov (Lyapunov, 1992), as for the equilibrium points, when all their eigenvalues of the Jacobian Matrix (J) are negative, they are the ESS. The eigenvalues of the Jacobian Matrix (J) for each equilibrium point are listed in Table 7, and ESS and corresponding conditions of each case are shown in Table 8.

The result is similar to that of Scenario 1. In this scenario, the ideal state is that foundation model providers adopt the PTM strategy, independent evaluators choose the MSA strategy and the public sector uses the CSS strategy, represented as $E_8'(1,1,1)$. However, based on the assumptions, this state could not be realised. Hence, with the target of trustworthy foundation model provision, the realisable and meaningful ESS is $E_5'(1,0,0)$, where the providers provide trustworthy services while the evaluators and public sector do not work rigorously. In this case, for foundation model providers, the key costs of providing untrustworthy services are higher than the deduction of the ecosystem benefits from the combined costs of supplying trustworthy offerings and conducting evaluation innovation alone. Furthermore, the key costs of adopting the PUM strategy include the direct costs *UMC*, indemnities for customers *I* and negative ecosystem effects *NE*. The major costs of supplying trustworthy offerings without other players engaging in evaluation innovation include direct costs *TMC* and overall evaluation innovation costs *EIC*. This condition is identical to that of Case 3 in the regulation scenario of self-assessment. Regardless of the strategy that other players employ, as long as the providers provide untrustworthy service, that is, the ESS are $E_1'(0,0,0)$, $E_2'(0,0,1)$ and $E_3'(0,1,0)$, independent assessment and market surveillance are complete failures.

4.2. Stability of the foundation model providers' strategies

4.2.1. Stability of the foundation model providers' strategies in scenario 1

According to the stability theorem of ordinary differential equations, when foundation model providers reach the ESS, it means that $F_1(x_1) = 0$ (let function (7) equal zero) and $F_1'(x_1) < 0$ (let the derivation of function (7) lower than zero).

Proposition 1. *For* $EII + r_mMR > 0$, *in the condition of* $y_1 > y^*$, *the stable strategy of foundation model providers would be PTM, and in the condition of* $y_1 < y^*$, *their stable strategy would be PUM, with the threshold* $y^* = \frac{TMC - UMC + EIC - I - NE - R}{EII + r_mMR}$ (*see* Appendix A *for the proof*).

The strategy evolution of foundation model providers would be affected by that of the public sector because the value of x_1 would be determined by the relationship between y_1 and y^*. The factors influencing the providers' strategy choice would cover those directly related to them as well as include all of the ecosystem-related factors and two public sector-related factors, namely, the investment in evaluation innovation *EII* and the rate of administrative fines r_m.

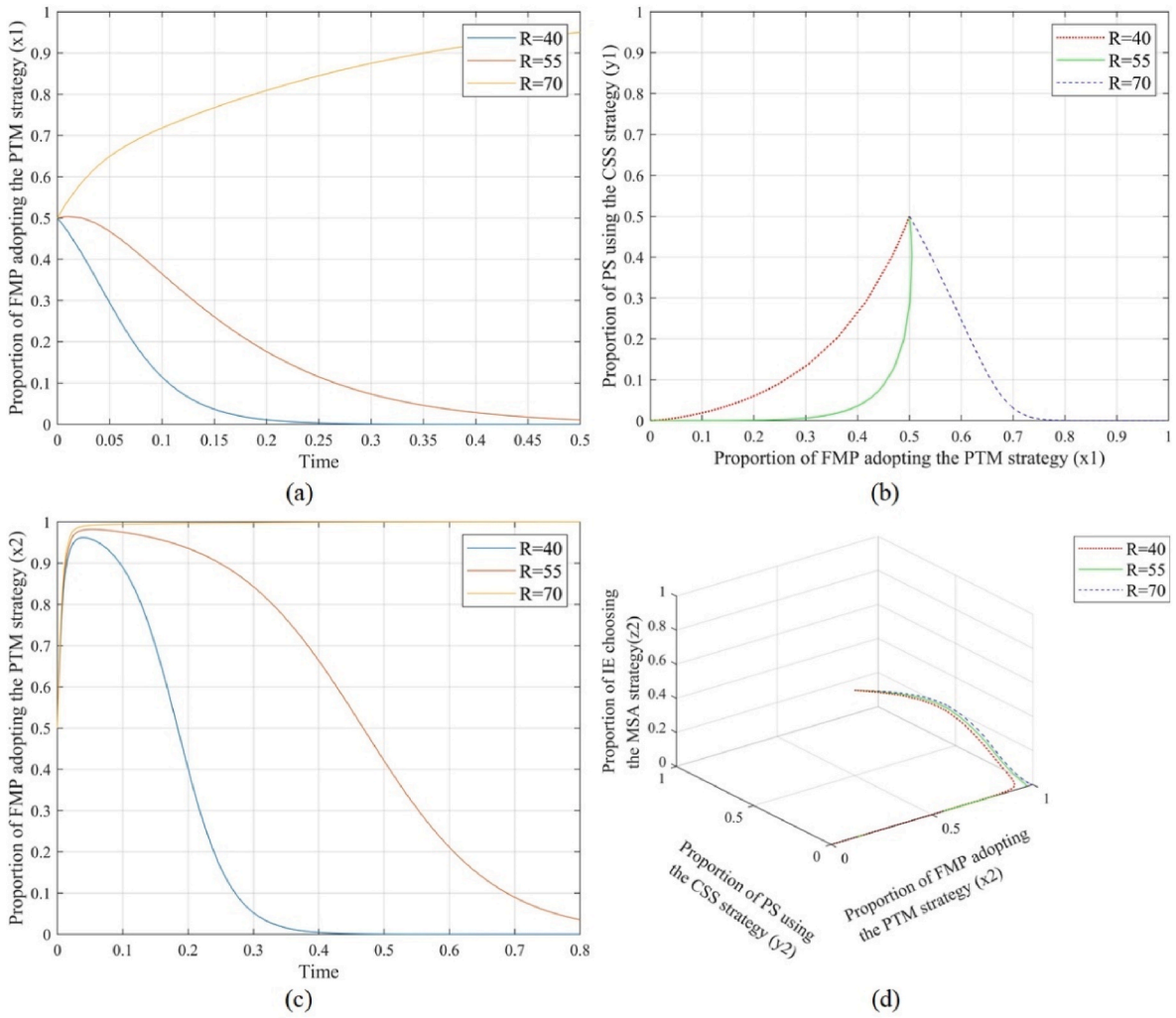


Fig. 9. Effects of R on the stable strategies of foundation model providers (a) and on the system ESS (b) in scenarios 1, and the corresponding effects in scenarios 2 (c–d). R denotes the ecosystem benefits brought by trustworthy models.

4.2.2. Stability of the foundation model providers' strategies in scenario 2

Likewise, the requirements for the ESS of foundation model providers are $F_2(x_2) = 0$ (let function (18) equal zero) and $F_2'(x_2) < 0$ (let the derivation of function (18) lower than zero).

Proposition 2.1. *For* $EII - z_2EII + r_mMR + z_2\lambda EII - z_2r_mMR > 0$, *in the condition of* $y_2 > y^{**}$, *the stable strategy of the providers is PTM, and in the condition of* $y_2 < y^{**}$, *it would be PUM, with the threshold* $y^{**} = \frac{EIC+TMC-UMC-I-NE-R-y_2[(1-\lambda)EIC-I-NE+MR-TC_1]}{EII+r_mMR-y_2[(1-\lambda)EII+r_mMR]}$ (*see* Appendix A *for the proof*).

Proposition 2.2. *In the condition* $(1-\lambda)EIC + MR - I - NE - TC_1 - y_2[(1-\lambda)EII + r_mMR] > 0$ *and* $z_2 > z^{**}$ *or that of* $(1-\lambda)EIC + MR - I - NE - TC_1 - y_2[(1-\lambda)EII + r_mMR] < 0$ *and* $z_2 < z^{**}$, *the stable strategy of foundation model providers would be PTM; in the condition of* $(1-\lambda)EIC + MR - I - NE - TC_1 - y_2[(1-\lambda)EII + r_mMR] > 0$ *and* $z_2 < z^{**}$ *or that of* $(1-\lambda)EIC + MR - I - NE - TC_1 - y_2[(1-\lambda)EII + r_mMR] < 0$ *and* $z_2 > z^{**}$, *their strategy would stabilise at PUM, with the threshold* $z^{**} = \frac{TMC-UMC+EIC-I-NE-R-y_2(EII+r_mMR)}{(1-\lambda)EIC+MR-I-NE-TC_1-y_2[(1-\lambda)EII+r_mMR]}$ (*see* Appendix A *for the proof*).

Proposition 2.3. *Due to* $z_2(y_2EII - EIC) < 0$, *in the condition of* $\lambda > \lambda^{**}$, *the providers' strategy choice would stabilise at PUM, and in the condition of* $\lambda < \lambda^{**}$, *their stable strategy would be PTM, with the threshold* $\lambda^{**} = \frac{y_2z_2(EII+r_mMR)-y_2(EII+r_mMR)+z_2(-EIC+I+NE-MR+TC_1)-I-NE-R+EIC+TMC-UMC}{z_2(y_2EII-EIC)}$ (*see* Appendix A *for the proof*).

In the regulation scenario of independent assessment, the strategy choices of the providers would be influenced by those of the other two groups, because the relationship between y_2 and y^{**} and between z_2 and z^{**} could determine the value of x_2. The factors affecting the strategy evolution of foundation model providers would include all those related to them and the ecosystems as well as those relevant to the public sector, the investment in evaluation innovation *EII* and the rate of administrative fines r_m.

4.3. Parameters' influence on foundation model providers' strategy choice

According to the stability analysis of foundation model providers' strategies in Section 4.2, we identify that the parameters *I*, *TMC*, *EIC*, *UMC*, *NE*, *R*, *EII*, r_m and *MR* in Scenarios 1 and 2, as well as TC_1 and λ newly in Scenario 2 would influence their strategy choice. Hence, we further analyse how these factors would make an impact.

In terms of foundation model providers' replicator dynamic equation in the regulation scenario of self-assessment (function (7)), the first-order partial derivatives of *I*, *TMC*, *EIC*, *UMC*, *NE*, *R*, *EII*, r_m, *MR* are $\frac{\partial F_1(x_1)}{\partial I} = x_1(1-x_1)$, $\frac{\partial F_1(x_1)}{\partial TMC} = -x_1(1-x_1)$, $\frac{\partial F_1(x_1)}{\partial EIC} = -x_1(1-x_1)$, $\frac{\partial F_1(x_1)}{\partial UMC} = x_1(1-x_1)$, $\frac{\partial F_1(x_1)}{\partial NE} = x_1(1-x_1)$, $\frac{\partial F_1(x_1)}{\partial R} = x_1(1-x_1)$, $\frac{\partial F_1(x_1)}{\partial EII} = x_1y_1(1-x_1)$, $\frac{\partial F_1(x_1)}{\partial r_m} = x_1y_1(1-x_1)MR$, $\frac{\partial F_1(x_1)}{\partial MR} = x_1y_1(1-x_1)r_m$, respectively, prompting

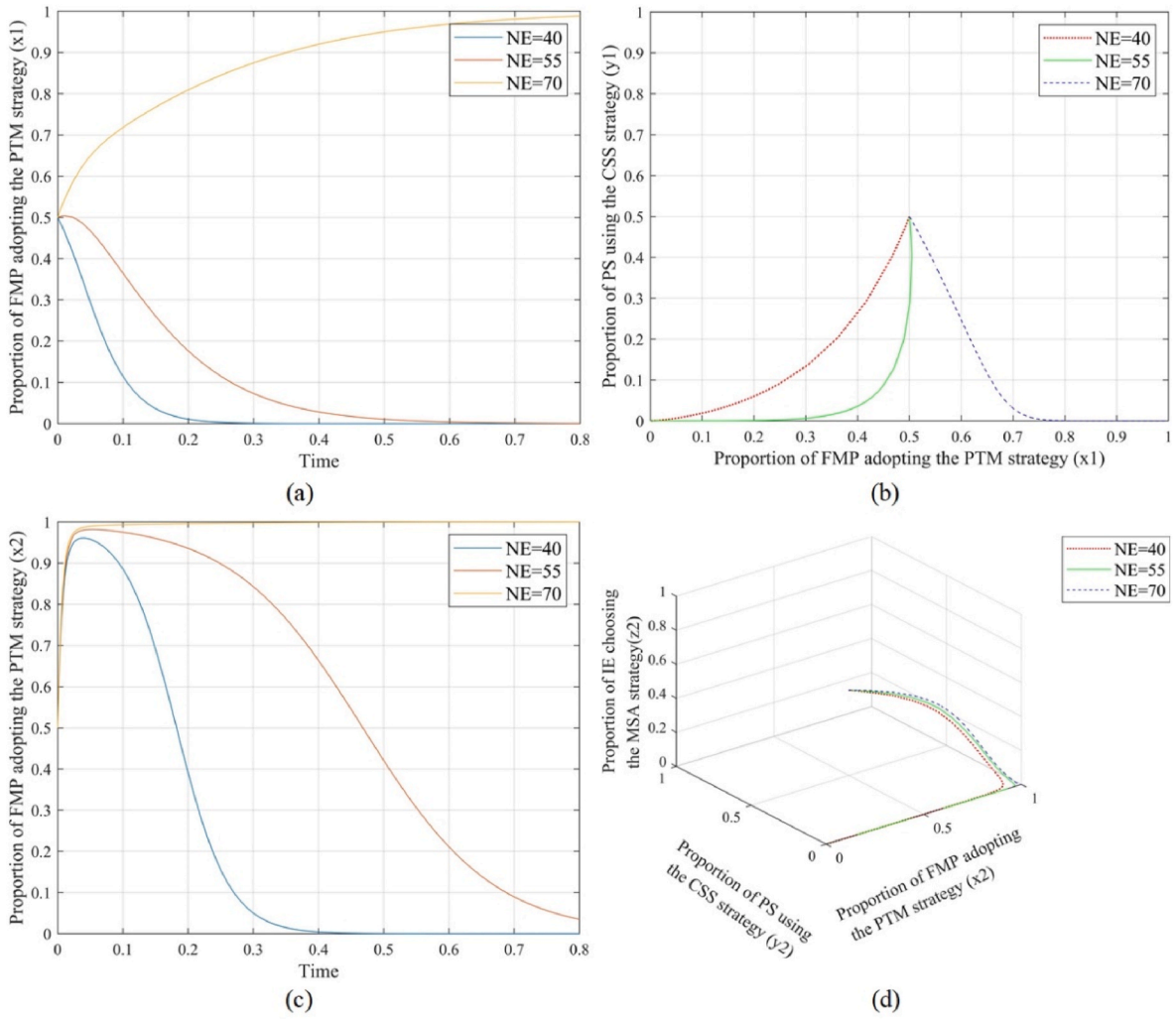


Fig. 10. Effects of *NE* on the stable strategies of foundation model providers (a) and on the system ESS (b) in scenarios 1, and the corresponding effects in scenarios 2 (c–d). *NE* denotes the negative ecosystem effects caused by untrustworthy models.

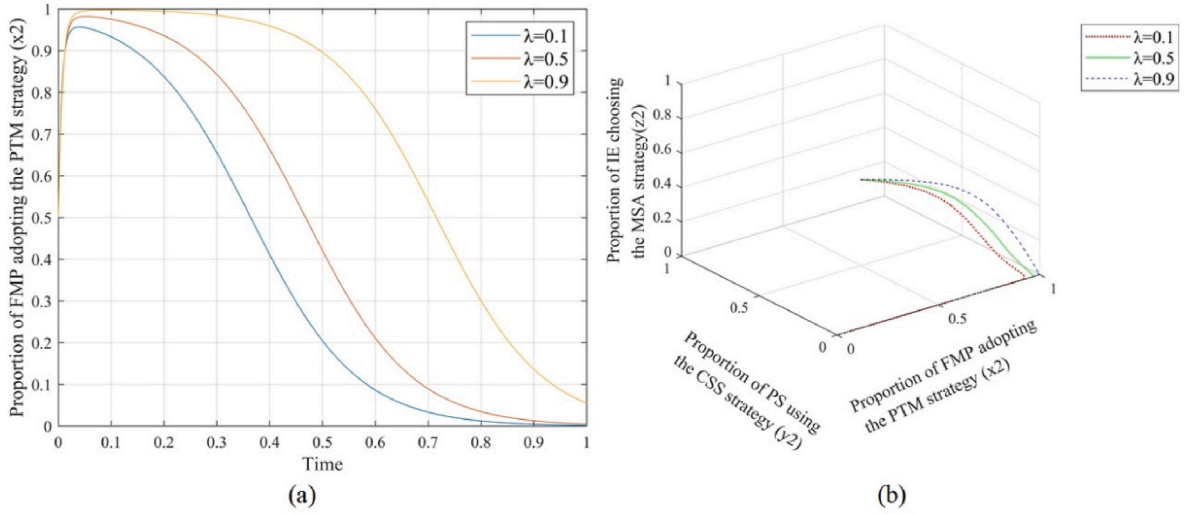


Fig. 11. Effects of λ on the stable strategies of foundation model providers (a) and on the system ESS (b) in Scenario 2. λ is the shared coefficient of evaluation innovation costs between foundation model providers and independent evaluators.

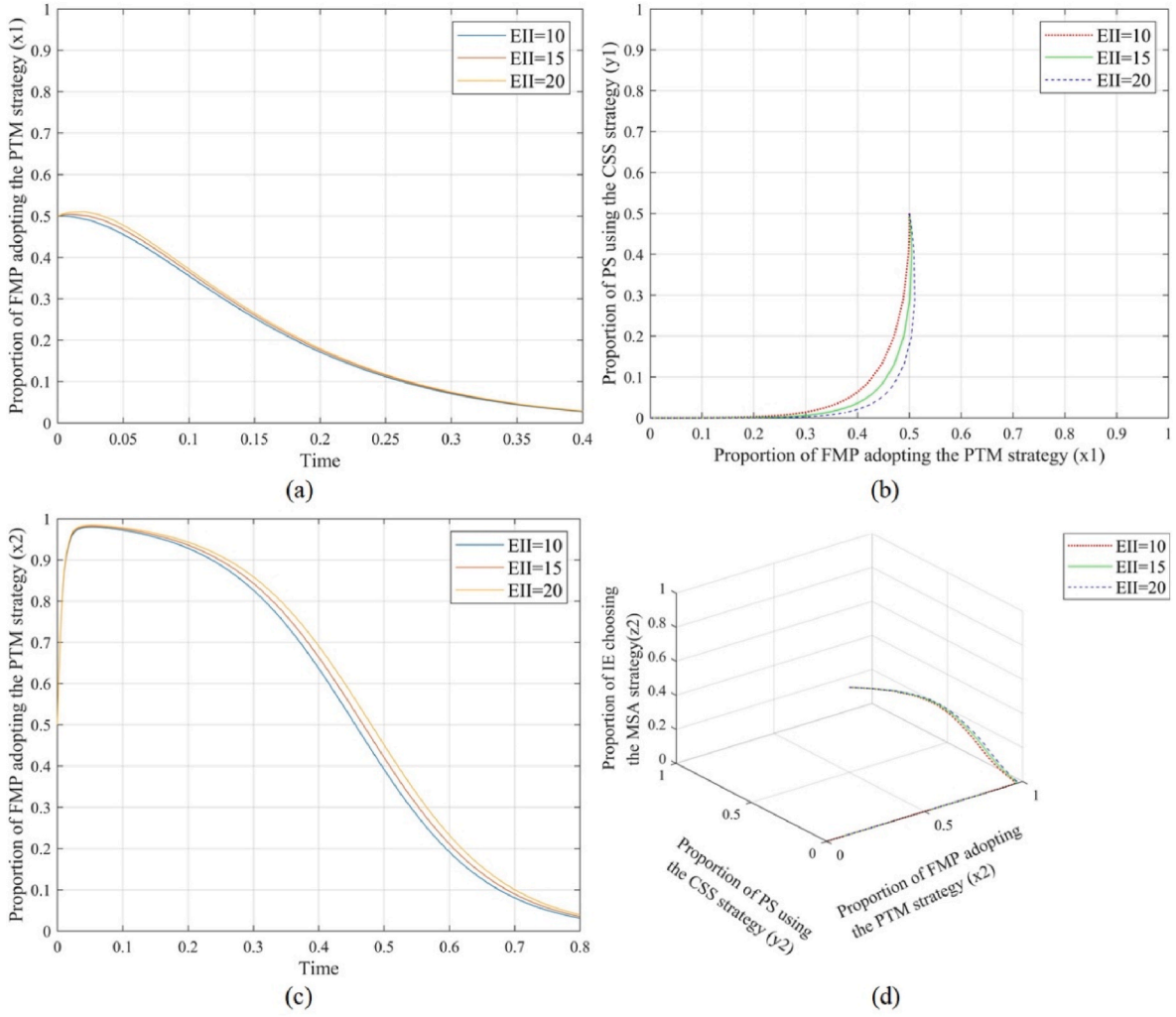


Fig. 12. Effects of *EII* on the stable strategies of foundation model providers (a) and on the system ESS (b) in scenarios 1, and the corresponding effects in scenarios 2 (c–d). *EII* denotes the public sector's investment in evaluation innovation.

Proposition 3. *I, UMC, NE, R, EII, r_m and MR would positively impact foundation model providers choosing the PTM strategy, while TMC and EIC would have negative effects on the provision of trustworthy service.*

According to the providers' replicator dynamic equation in the regulation scenario of independent assessment (function (18)), the first-order partial derivatives of *I*, *TMC*, *EIC*, *UMC*, *NE*, *R*, *EII*, r_m and *MR* that are the same parameters as those in Scenario 1 and TC_1 and λ in Scenario 2, are $\frac{\partial F_2(x_2)}{\partial I} = x_2(1-x_2)(1-z_2)$, $\frac{\partial F_2(x_2)}{\partial TMC} = -x_2(1-x_2)$, $\frac{\partial F_2(x_2)}{\partial EIC} = -x_2(1-x_2)(1-z_2+\lambda z_2)$, $\frac{\partial F_2(x_2)}{\partial UMC} = x_2(1-x_2)$, $\frac{\partial F_2(x_2)}{\partial NE} = x_2(1-x_2)(1-z_2)$, $\frac{\partial F_2(x_2)}{\partial R} = x_2(1-x_2)$, $\frac{\partial F_2(x_2)}{\partial EII} = x_2(1-x_2)[y_2-(1-\lambda)y_2z_2]$, $\frac{\partial F_2(x_2)}{\partial r_m} = x_2(1-x_2)(1-z_2)y_2MR$, $\frac{\partial F_2(x_2)}{\partial MR} = x_2(1-x_2)[z_2+y_2(1-z_2)r_m]$, $\frac{\partial F_2(x_2)}{\partial TC_1} = -x_2z_2(1-x_2)$, $\frac{\partial F_2(x_2)}{\partial \lambda} = -x_2(1-x_2)z_2(EIC-y_2EII)$, respectively, prompting

Proposition 7. *I, UMC, NE, R, EII, r_m and MR would have a positive correlation with the probability of the providers using the PTM strategy, but on the contrary, TMC, EIC, TC_1 and λ negatively affect their choice of the PTM strategy.*

Thus, the same factors in these two scenarios would have highly similar effects on the strategy evolution of the providers, and the newly added factors TC_1 and λ in Scenario 2 would negatively affect the provision of trustworthy models. However, we find that the factors *EII*, r_m, *MR*, TC_1 or λ could affect the strategy choices of foundation model providers. Meanwhile, according to the findings in Section 4.1, they could not make a difference in achieving the system ESS, $E_3(1,0)$ or $E_5'(1,0,0)$.

5. Numerical analysis

In this section, using MATLAB R2020a, we numerically simulate the evolutionary game models of the two scenarios and illustrate whether the initial choices of these actors could affect the evolution results and how the factors discussed in the model analysis could influence the selections of foundation model providers and the system ESS. The values of all parameters are provided under the requirements of the assumptions, and the initial ones are listed in Table 9 under the conditions of Cases 1 and 4. This is because in the early stage of foundation model development, the models have not yet achieved trustworthiness, and market surveillance and independent assessment could not be strict. Moreover, the initial value of the rate of the administrative fines r_m is given by referring to the related regulation in the EU's AI Act (European Commission, 2024).

5.1. The effects of the major actors' strategy choice

The initial percentages of foundation model providers adopting the PTM strategy, x_1 and x_2 in Scenarios 1 and 2, are varied at 0.1, 0.5 and 0.9 to ascertain whether their initial choice could influence the evolu-

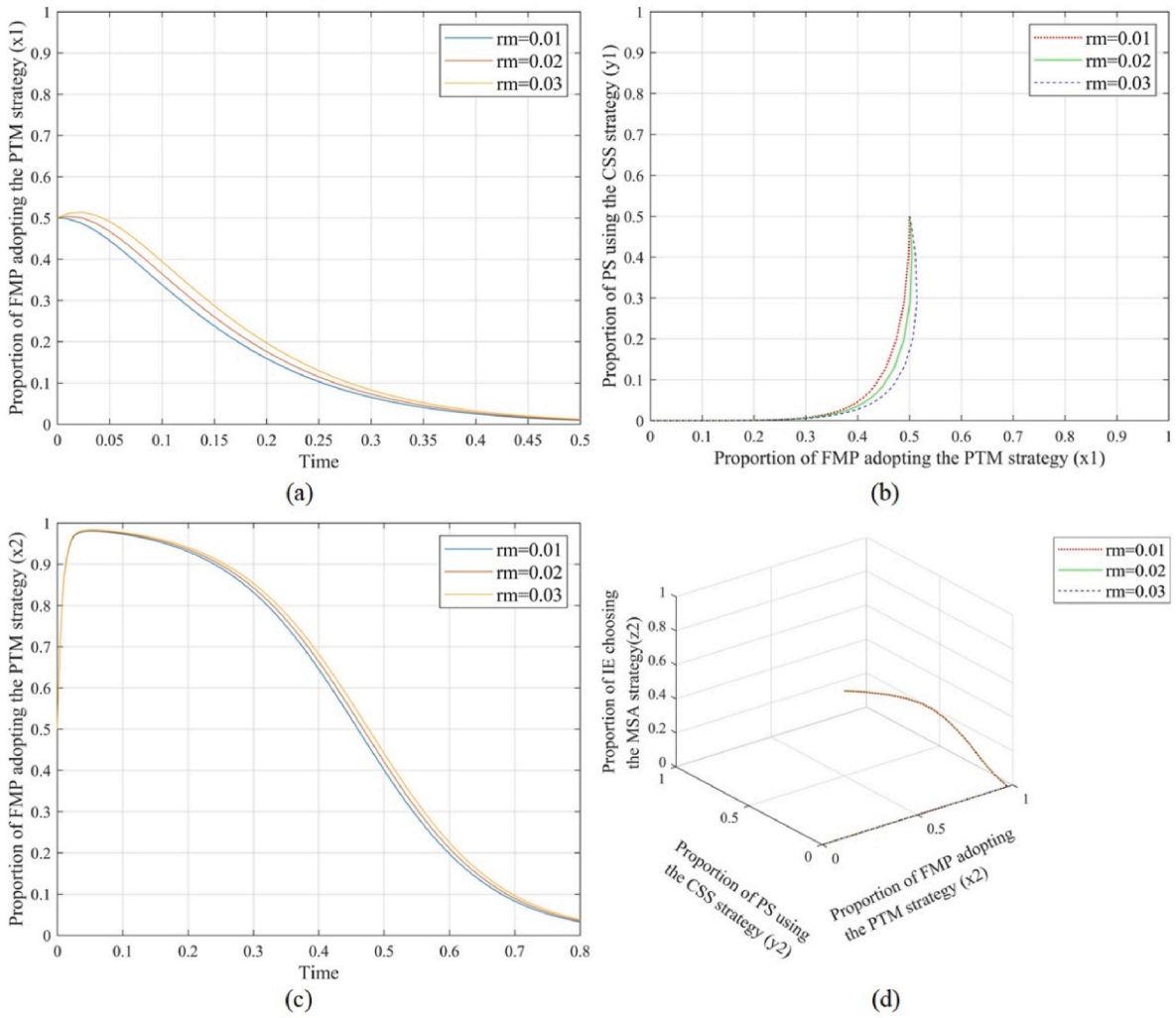


Fig. 13. Effects of r_m on the stable strategies of foundation model providers (a) and on the system ESS (b) in scenarios 1, and the corresponding effects in scenarios 2 (c–d). r_m is the rate of administrative fines for the providers providing untrustworthy models.

tion results (Fig. 2(a and b)). Similarly, the initial proportion of the public sector using the CSS strategy, y_1 and y_2 in Scenarios 1 and 2 (Fig. 2(c and d)), as well as the percentages of independent evaluators employing the MSA strategy z_2 are all set at 0.1, 0.5 and 0.9 (Fig. 2(e)). Fig. 2(a–e) demonstrates that regardless of the initial values, the ESS of the systems in scenarios 1 and 2 converge to $E_1(0,0)$ and $E_1'(0,0,0)$, respectively. These findings differ from those in Section 4.2, 'Stability of the Foundation Model Providers' Strategies' but align with the results of Cases 1 and 4. The discrepancy arises because although the initial choices of players can affect the evolutionary process, they do not alter the final equilibrium under the given conditions. Therefore, the initial strategic choices of all actors do not affect the ultimate evolution results, and their decision-making is primarily determined by key influencing factors.

5.2. The effects of the provider-related factors

The value of the direct costs of providing trustworthy foundation models is varied at 150, 200 and 250 with other parameters unchanged. The stable strategy of foundation model providers changes from the PTM strategy to the PUM strategy in both scenarios. The system ESS converts from $E_3(1,0)$ to $E_1(0,0)$ in Scenario 1 (Fig. 3(a and b); similarly, changes also occur in Scenario 2 with the ESS varying from $E_5'(1,0,0)$ to $E_1'(0,0,0)$ (Fig. 3(c and d)), which is consistent with the results in Section 4. By contrast, there is an opposite change with the variance of *UMC* and *I*. When the costs of offering untrustworthy services *UMC* are set at 90, 140 and 190 and the indemnities given to consumers *I* are set at 20, 40 and 60 with other parameters remaining, the evolutionary result of the providers' behaviour changes from adopting the PUM strategy to using the PTM strategy in both scenarios and the ESS transforms from $E_1(0,0)$ to $E_3(1,0)$ in Scenario 1, and from $E_1'(0,0,0)$ to $E_5'(1,0,0)$ (Fig. 4(a–d) and Fig. 5(a–d)). The related analyses in Section 4 are also confirmed by this.

When other parameters are constant, the incomes of foundation model provision *MR* are varied at 490, 550 and 610, which do not make an obvious difference on the providers' choice or the system ESS in the two scenarios (Fig. 6(a–d)). According to the model analyses in Section 4, it is not the key factor that could change the ESS from $E_1(0,0)$ to $E_3(1,0)$ in Scenario 1 or from $E_1'(0,0,0)$ to $E_5'(1,0,0)$ in Scenario 2. However, its increase would slightly encourage choosing the PTM strategy, as demonstrated clearly in Fig. 6(a–d). Similarly, the time costs of launch delay TC_1 are set at 30, 50 and 70 with other parameters constant in Scenario 2, which would neither totally change the providers' choice nor the system ESS (Fig. 7(a and b)).

5.3. The effects of the ecosystem-related factors

Fig. 8(a–d) shows that when other parameters are constant and the overall costs of evaluation innovation *EIC* are varied at 80, 100 and 120, the final choice of foundation model providers changes from the PTM to

PUM strategy, and the ESS converts from $E_3(1,0)$ to $E_1(0,0)$ in Scenario 1 and from $E_5'(1,0,0)$ to $E_1'(0,0,0)$ in Scenario 2. Meanwhile, the values of both the ecosystem benefits from trustworthy models R and the ecosystem negative effects of untrustworthy ones NE are set at 40, 50 and 70. These parameter changes would alter the evolution strategy of the providers, shifting from PUM to PTM. Correspondingly, the ESS transitions from $E_1(0,0)$ to $E_3(1,0)$ in Scenario 1, and from $E_1'(0,0,0)$ to $E_5'(1,0,0)$ in Scenario 2, as illustrated in Fig. 9(a–d) and Fig. 10(a–d), respectively. As for the shared coefficient of evaluation innovation costs between foundation model providers and independent evaluators λ, when it is set at 0.1, 0.5 and 0.9, the evolution results of the providers' strategy choice and the system ESS would be unchanged (Fig. 11(a and b)).

5.4. The effects of the public sector-related factors

Fig. 12(a–d) shows that when the public sector's investment in evaluation innovation EII is varied at 10, 15 and 20 with other parameters fixed, the evolution results of the providers' strategy choice and the system ESS would remain constant. Furthermore, similar situations would occur when the rate of administrative fines for the providers' non-compliance behaviour r_m is set at 0.01, 0.02 and 0.03 (Fig. 13(a–d)). These findings further support the results in Section 4 that these two factors would positively affect foundation model providers choosing the PTM strategy but could not have any influence on the system ESS in both scenarios.

Based on the numerical analyses, three major findings emerge. First, the initial choices of each actor would not influence the providers' final stable strategy or the system ESS; only the key factors could make a difference. Second, the key factors that could transform the system ESS would fall into two types: provider-related and ecosystem-related. The former includes the direct costs for the providers providing trustworthy models, the direct costs of offering untrustworthy service and the indemnities given to consumers. The latter includes the overall costs of evaluation innovation, the ecosystem benefits brought by trustworthy models and the negative ecosystem effects caused by untrustworthy service. This finding is consistent with the conditions of Cases 3 and 7 in Section 4.1. Third, although some factors could influence the providers' decision-making to some degree, they could not determine the result of system evolution. These include public-sector-related factors such as the investment in evaluation innovation and the rate of administrative fines, provider-related factors such as the incomes of foundation models and the time costs of launch delay for independent assessment, and the ecosystem-related factor—the shared coefficient of evaluation innovation costs between foundation model providers and independent evaluators. The simulated charts of these factors visually demonstrate their limited effects, which also correspond with the results of the model analyses in Section 4.

6. Discussion and conclusion

Based on the model and numerical analyses, we identified three key findings and discussed their implications for developing trustworthy foundation models. First, the attainment of trustworthiness is likely contingent upon several provider- and ecosystem-related factors rather than solely by strict ex ante conformity assessment regulations. The provider-related factors include the direct costs incurred by the providers in offering trustworthy models and untrustworthy services as well as the indemnities given to consumers. The ecosystem-related factors include the overall costs of evaluation innovation, ecosystem benefits brought by trustworthy models and negative ecosystem effects caused by untrustworthy services.

Ecosystem impacts, such as the benefits of trustworthy models and negative effects of untrustworthy ones, play a crucial role in shaping providers' decisions. Therefore, entrepreneurs and developers should adopt a forward-thinking approach, or 'ecosystem thinking', recognising that foundation models have long-term implications for the broader innovation ecosystem. This perspective underscores the importance of embedding trustworthiness into model design and training, as all actors within the ecosystem share a collective fate with foundation models and society is affected by them. Although these models drive innovation and progress, they pose risks, making trustworthiness a fundamental prerequisite for sustainable enterprise growth. This aligns with Powers et al.'s view that a much broader environment must be considered to ensure AI trustworthiness (Powers et al., 2023).

Furthermore, the costs of offering untrustworthy services and indemnities to consumers can be considered market-driven pressures. If consumers prioritise trustworthiness, the expenses associated with falsifying evaluation documents and providing compensation are likely to rise. In this context, influential downstream developers and other consumers play a crucial role in encouraging providers to develop trustworthy foundation models. Given market demand and ecosystem impacts, trustworthiness should be considered not a liability but a competitive advantage for providers (Küspert et al., 2023). Stakeholders in foundation model development must recognise the strategic value of prioritising trustworthiness to align with evolving market expectations and ecosystem dynamics.

Although market-driven incentives can play an important role in promoting trustworthiness, self-regulation alone has significant limitations, as evidenced in various domains such as consumer privacy protection. Research has shown that without legal enforcement, market forces and industry-led self-regulation are often insufficient to mitigate risks. For example, according to Floridi (2022), self-regulation has failed to prevent issues such as online disinformation, underscoring the need for stronger legal oversight in AI governance (Floridi, 2021). Similarly, Mazurek and Małagocka (2019) recognised the strategic value of self-regulation but emphasised that governments must take a more active role in addressing the ethical, privacy and societal challenges posed by AI (Mazurek & Małagocka, 2019). Moreover, de Marcellis-Warin et al. (2022) underscored that allowing companies to assess the risks of their algorithms is inadequate for ensuring consumer privacy and protection (de Marcellis-Warin et al., 2022). Therefore, rather than relying solely on market mechanisms, a more effective approach would involve a hybrid regulatory framework that combines legal mandates with industry-driven incentives and ecosystem considerations. Such a balanced framework would provide stronger safeguards against potential harm, foster innovation and build trust in foundation models.

Second, ex ante conformity assessment regulations may not play a central role in incentivising foundation model providers to deliver trustworthy offerings. This is supported by two findings: first, even in the best-case regulatory scenarios, only providers tend to adopt a proactive strategy, while the public sector are inactive; second, public sector-related factors do not influence providers' final decisions or the system's evolution outcomes. However, market surveillance remains necessary. While it may not drive providers' decisions directly, punitive measures can act as deterrents and public sector investment can help ease the burden of evaluation innovation. Additionally, the public sector should actively safeguard the rights and interests of downstream providers and other stakeholders, recognising existing power imbalances.

When faced with a large benefit–cost gap, providers are unlikely to voluntarily adhere to trustworthiness principles. In such cases, strict regulations become essential. Wirtz et al. (2023) introduced the concept of corporate digital responsibility (CDR) and argued that most firms are unlikely to implement effective CDR practices because the costs often exceed the benefits (Wirtz et al., 2023). Kunz and Wirtz (2023) applied CDR to foundation models, demonstrating that several companies prioritise financial gains over responsible AI practices, despite the associated risks—especially as AI technologies amplify the value of data and exploit its potential (Kunz & Wirtz, 2024). A clear example is how companies such as Meta and Google actively resist privacy and AI

regulations. Thus, when market forces fail to encourage ethical behaviour and actively incentivise unethical practices, governments must enforce strict regulations and legal frameworks to protect consumers and ensure responsible AI development. Furthermore, more effective regulatory mechanisms must be established to address market failures and strengthen trustworthiness in foundation models.

Third, the regulation of independent assessment may not produce ideal effects as no factors associated with this regulation exert any influence on the choices made by providers or the evolution of the system's ESS. However, considering the current landscape, the evaluation of trustworthiness principles through technical means and their integration into model development pose substantial challenges, necessitating the development of independent evaluation tools and methods. In this context, the creation of independent evaluators, acting as developers of third-party auditing tools and methods, becomes pivotal. Thus, these evaluators can contribute to the shared costs of overall evaluation innovation, providing a compelling incentive for providers to furnish trustworthy models. Consequently, within the innovation ecosystems of foundation models, independent evaluators emerge as crucial actors, collaborating with providers to audit and improve model trustworthiness rather than merely serving as components of market surveillance. This underscores the need to reconsider the elements and structure of future foundation-model-based innovation ecosystems. Furthermore, a significant disparity persists between AI ethics principles and the actual design of AI systems. Bridging this gap and ensuring the ethical governance of AI–an essential component of trustworthiness–requires the active involvement of both AI practitioners and external auditors (Morley et al., 2021).

In conclusion, the profound influence and rapid evolution of foundation models have raised widespread concerns regarding their trustworthiness. It is essential to consider whether potential regulatory regimes could be sufficiently effective and understand its reasons. One important approach to managing high-risk AI systems is ex ante conformity assessment regulations, which can also be used to govern foundation models. This study constructs game models to predict the possible effects of these regulatory methods and analyse the key factors influencing the behaviour of foundation model providers. The goal is to enrich policy discussions and offer valuable references for future policy formulation.

We establish two hypothetical scenarios—self-assessment and independent conformity assessment—infused with an innovation ecosystem perspective and analysed through an evolutionary game approach. Our findings indicate that the two regulatory approaches of conformity assessment do not exert distinct influences. Instead, the attainment of trustworthiness is shaped primarily by factors such as market requirements, ecosystem impacts, the overall costs of evaluation innovation and the costs associated with the innovation and operation of trustworthy models. Thus, we conclude with a discussion of the implications derived from our research.

It is important to acknowledge a limitation of this study. Trustworthiness in foundation models is a broad concept, and we do not examine the specific regulatory measures targeting particular aspects of trustworthiness. Therefore, future research could explore tailored regulations that address these specific dimensions. Further discussion is needed on a hybrid regulatory framework that balances legal mandates with industry-driven incentives and ecosystem impacts to ensure accountability and innovation.

CRediT authorship contribution statement

Xiaoxu Zhang: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Conceptualization. **Wenyong Zhou:** Methodology, Conceptualization. **Wen Hu:** Visualization, Software. **Shenghan Zhou:** Writing – original draft, Funding acquisition, Writing – review & editing. **Xiaoqian Hu:** Supervision, Funding acquisition. **Linchao Yang:** Writing – review & editing, Supervision, Methodology, Conceptualization, Funding acquisition.

Declaration of generative AI and AI-assisted technologies in the writing process

Statement: During the preparation of this work, the authors used ChatGPT-3.5 and ChatGPT-4o to improve the readability and language of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that there are no conflicts of interest related to this work.

Acknowledgement

This research was supported by the National Natural Science Foundation of China (Grant Nos. 72371013, 72101164, and 72401097), and the Fundamental Research Funds for the Central Universities (YWF-23-L-933).

Appendix

Proof for Propositions

Proof of Proposition 1.

Let $F_1(x_1) = \frac{dx_1}{dt} = x_1\left(U_{p11} - U_{p1}\right) = -x_1(x_1 - 1)(I - TMC - EIC + UMC + NE + R + y_1 EII + y_1 r_m MR) = 0$ and $F_1'(x_1) = -(2x_1 - 1)(I - TMC - EIC + UMC + NE + R + y_1 EII + y_1 r_m MR) < 0$. We make $G(y) = y_1(EII + r_m MR) + I + NE + R + UMC - TMC - EIC$ that is an increasing function for y_1, and when $y_1 > y^*$, $G(y_1) > G(y^*)$, $F_1(x)|_{x_1=1} = 0$, $F_1'(x)|_{x_1=1} < 0$, here, $x_1 = 1$ is the stable evolutionary strategy for foundation model providers. Whereas, when $y_1 < y^*$, $G(y_1) < G(y^*)$, $F(x)|_{x_1=0} = 0$, $F_1'(x)|_{x_1=0} < 0$, here, $x_1 = 0$ is the stable evolutionary strategy for foundation model providers. If $y_1 = y^*$, $G(y_1) = G(y^*)$, $F(x) = 0$ and $F_1'(x) = 0$, in this case, we cannot ensure a stable strategy because all the strategies of foundation model providers are stable.

Proof of Proposition 2.1.

Let

$F_2(x_2) = \frac{dx_2}{dt} = x_2\left(U_{p21} - U_{p2}\right) = -x_2(x_2 - 1)(I - TMC - EIC + UMC + NE + R + y_2 EII + z_2 EIC - z_2 I - z_2 NE + z_2 MR - z_2 TC_1 - z_2 \lambda EIC - y_2 z_2 EII + y_2 r_m MR - + y_2 z_2 \lambda EII - y_2 z_2 r_m MR) = 0$ and $F_2'(x_2) = -(2x_2 - 1)(I - TMC - EIC + UMC + NE + R + y_2 EII + z_2 EIC - z_2 I - z_2 NE + z_2 MR - z_2 TC_1 - z_2 \lambda EIC - y_2 z_2 EII + y_2 r_m MR + y_2 z_2 \lambda EII - y_2 z_2 r_m MR) < 0$. We make $H(y) = y_2\{EII + r_m MR - z_2[(1 - \lambda)EII + r_m MR]\} + z_2[(1 - \lambda)EIC - I - NE + MR - TC_1] + I -$

$TMC - EIC + UMC + NE + R$ and it is an increasing function of y_2. When $y_2 > y^{**}$, $H(y_2) > H(y^{**})$, $F_2(x)|_{x_2=1} = 0$, $F_2'(x)|_{x_2=1} < 0$, here, $x_2 = 1$ is the ESS of foundation model providers. When $y_2 < y^{**}$, $H(y_2) < H(y^{**})$, $F_2(x)|_{x_2=0} = 0$, $F_2'(x)|_{x_2=0} < 0$, here, $x_2 = 0$ is the ESS of foundation model providers. When $y_2 = y^{**}$, $H(y_2) = H(y^{**})$, $F_2(x) = 0$ and $F_2'(x) = 0$, all the foundation model providers' strategies would be stable, so the ESS is uncertain.

Proof of Proposition 2.2

Let

$F_2(x_2) = \frac{dx_2}{dt} = x_2(U_{p21} - U_{p2}) = -x_2(x_2 - 1)(I - TMC - EIC + UMC + NE + R + y_2EII + z_2EIC - z_2I - z_2NE + z_2MR - z_2TC_1 - z_2\lambda EIC - y_2z_2EII + y_2r_mMR - + y_2z_2\lambda EII - y_2z_2r_mMR) = 0$ and $F_2'(x_2) = -(2x_2 - 1)(I - TMC - EIC + UMC + NE + R + y_2EII + z_2EIC - z_2I - z_2NE + z_2MR - z_2TC_1 - z_2\lambda EIC - y_2z_2EII + y_2r_mMR + y_2z_2\lambda EII - y_2z_2r_mMR) < 0$. We make $H(z) = z_2\{(1 - \lambda)EIC + MR - I - NE - TC_1 - y_2[(1 - \lambda)EII + r_mMR]\} + y_2(EII + r_mMR) + I + NE + R + UMC - TMC - EIC$, and when $(1 - \lambda)EIC + MR - I - NE - TC_1 - y_2[(1 - \lambda)EII + r_mMR] > 0$, $H(z)$ is an increasing function for z_2. When $z_2 > z^{**}$, $H(z_2) > H(z^{**})$, $F_2(x)|_{x_2=1} = 0$, $F_2'(x)|_{x_2=1} < 0$, here, the ESS of foundation model providers is $x_2 = 1$. When $z_2 < z^{**}$, $H(z_2) < H(z^{**})$, $F_2(x)|_{x_2=0} = 0$, $F_2'(x)|_{x_2=0} < 0$, here, the ESS of foundation model providers is $x_2 = 0$. When $z_2 = z^{**}$, $H(z_2) = H(z^{**})$, $F_2(x) = 0$ and $F_2'(x) = 0$, in this case, all the strategies of foundation model providers would be stable, so the ESS is uncertain. Furthermore, when $(1 - \lambda)EIC + MR - I - NE - TC_1 - y_2[(1 - \lambda)EII + r_mMR] < 0$, $H(z)$ is a decreasing function for z_2, the proposition could be proved similarly.

Proof of Proposition 2.3.

Let

$F_2(x_2) = \frac{dx_2}{dt} = x_2(U_{p21} - U_{p2}) = -x_2(x_2 - 1)(I - TMC - EIC + UMC + NE + R + y_2EII + z_2EIC - z_2I - z_2NE + z_2MR - z_2TC_1 - z_2\lambda EIC - y_2z_2EII + y_2r_mMR - + y_2z_2\lambda EII - y_2z_2r_mMR) = 0$ and $F_2'(x_2) = -(2x_2 - 1)(I - TMC - EIC + UMC + NE + R + y_2EII + z_2EIC - z_2I - z_2NE + z_2MR - z_2TC_1 - z_2\lambda EIC - y_2z_2EII + y_2r_mMR + y_2z_2\lambda EII - y_2z_2r_mMR) < 0$. We make $\varphi(\lambda) = -\lambda z_2(EIC - y_2EII) - y_2z_2(EII + r_mMR) + y_2(EII + r_mMR) + z_2(EIC - I - NE + MR - TC_1) + I - EIC + NE + R - TMC + UMC$ and it is a decreasing function of λ. When $\lambda > \lambda^{**}$, $\varphi(\lambda) < \varphi(\lambda^{**})$, $F_2(x)|_{x_2=0} = 0$, $F_2'(x)|_{x_2=0} < 0$, then $x_2 = 0$ is the ESS for foundation model providers. When $\lambda < \lambda^{**}$, $\varphi(\lambda) > \varphi(\lambda^{**})$, $F_2(x)|_{x_2=1} = 0$, $F_2'(x)|_{x_2=1} < 0$, then $x_2 = 1$ is the ESS for foundation model providers. If $\lambda = \lambda^{**}$, $\varphi(\lambda) = \varphi(\lambda^{**})$, $F_2(x) = 0$ and $F_2'(x) = 0$, in this case, the ESS of foundation model providers is uncertain because the strategies for $x_2 \in [0, 1]$ are stable.

Data availability

No data was used for the research described in the article.

References

Adner, R. (2006). Match your innovation strategy to your innovation ecosystem. *Harvard Business Review, 84*(4), 98. + https://hbr.org/2006/04/match-your-innovation-strategy-to-your-innovation-ecosystem.

Alalawi, Z., Bova, P., Cimpeanu, T., Stefano, A. D., Duong, M. H., Domingos, E. F., Anh, H. T., Krellner, M., Ogbo, B., Powers, S. T., & Zimmaro, F. (2024). Trust AI regulation? Discerning users are vital to build trust and effective AI regulation. *arXiv*. https://doi.org/10.48550/arXiv.2403.09510

Arenal, A., Armuña, C., Feijoo, C., Ramos, S., Xu, Z., & Moreno, A. (2020). Innovation ecosystems theory revisited: The case of artificial intelligence in China. *Telecommunications Policy, 44*(6), Article 101960. https://doi.org/10.1016/j.telpol.2020.101960

Avin, S., Belfield, H., Brundage, M., Krueger, G., Wang, J., Weller, A., Anderljung, M., Krawczuk, I., Krueger, D., Lebensold, J., Maharaj, T., & Zilberman, N. (2021). Filling gaps in trustworthy development of AI. *Science, 374*(6573), 1327–1329. https://www.science.org/doi/10.1126/science.abi7176.

Bengio, Y., Hinton, G., Yao, A., Song, D., Abbeel, P., Darrell, T., Harari, Y. N., Zhang, Y.-Q., Xue, L., Shalev-Shwartz, S., Hadfield, G., Clune, J., Maharaj, T., Hutter, F., Baydin, A. G., McIlraith, S., Gao, Q., Acharya, A., Krueger, D., … Mindermann, S. (2024). Managing extreme AI risks amid rapid progress. *Science, 384*(6698), 842–845. https://www.science.org/doi/10.1126/science.adn0117.

Bommasani, R., Hudson, D. A., Adeli, E., Altman, R., Arora, S., Arx, S.v., Bernstein, M. S., Bohg, J., Bosselut, A., Brunskill, E., Brynjolfsson, E., Buch, S., Card, D., Castellon, R., Chatterji, N. S., Chen, A. S., Creel, K. A., Davis, J., Demszky, D., … Liang, P. (2022). On the opportunities and risks of foundation models. *arXiv*. https://doi.org/10.48550/arXiv.2108.07258

Bommasani, R., Klyman, K., Zhang, D., & Liang, P. (2023). Do foundation model providers comply with the draft EU AI act?. https://crfm.stanford.edu/2023/06/15/eu-ai-act.html. (Accessed 27 November 2023).

Bova, P., Stefano, A. D., & Anh, H. T. (2023). Both eyes open: Vigilant incentives help regulatory markets improve AI safety. *arXiv*. https://doi.org/10.48550/arXiv.2303.03174

Brundage, M., Avin, S., Wang, J., Belfield, H., Krueger, G., Hadfield, G., Khlaaf, H., Yang, J., Toner, H., Fong, R., Maharaj, T., Koh, P. W., Hooker, S., Leung, J., Trask, A., Bluemke, E., Lebensold, J., O'Keefe, C., Koren, M., … Anderljung, M. (2020). Toward trustworthy AI development: Mechanisms for supporting verifiable claims. *arXiv*. https://doi.org/10.48550/arXiv.2004.07213

Cammarota, R., Schunter, M., Rajan, A., Boemer, F., Kiss, Á., Treiber, A., Weinert, C., Schneider, T., Stapf, E., Sadeghi, A.-R., Demmler, D., Stock, J., Chen, H., Umar Hussain, S., Riazi, S., Koushanfar, F., Gupta, S., Simunic Rosing, T., Chaudhuri, K., … Norton, P. (2020). Trustworthy AI inference systems: An industry research view. *arXiv*. https://doi.org/10.48550/arXiv.2008.04449

de Marcellis-Warin, N., Marty, F., Thelisson, E., & Warin, T. (2022). Artificial intelligence and consumer manipulations: From consumer's counter algorithms to firm's self-regulation tools. *AI and Ethics, 2*(2), 259–268. https://doi.org/10.1007/s43681-022-00149-5

Dedehayir, O., Makinen, S. J., & Ortt, J. R. (2018). Roles during innovation ecosystem genesis: A literature review. *Technological Forecasting and Social Change, 136*, 18–29. https://doi.org/10.1016/j.techfore.2016.11.028

Díaz-Rodríguez, N., Del Ser, J., Coeckelbergh, M., López de Prado, M., Herrera-Viedma, E., & Herrera, F. (2023). Connecting the dots in trustworthy Artificial Intelligence: From AI principles, ethics, and key requirements to responsible AI systems and regulation. *Information Fusion, 99*, Article 101896. https://doi.org/10.1016/j.inffus.2023.101896

Encarnaçao, S., Santos, F. P., Santos, F. C., Blass, V., Pacheco, J. M., & Portugali, J. (2018). Paths to the adoption of electric vehicles: An evolutionary game theoretical approach. *Transportation Research Part B: Methodological, 113*, 24–33. https://doi.org/10.1016/j.trb.2018.05.002

European Commission. (2019a). Communication: Building trust in human centric artificial intelligence. https://digital-strategy.ec.europa.eu/en/library/communication-building-trust-human-centric-artificial-intelligence. (Accessed 13 November 2023).

European Commission. (2019b). Ethics guidelines for trustworthy AI. https://digital-strategy.ec.europa.eu/en/library/ethics-guidelines-trustworthy-ai. (Accessed 13 November 2023).

European Commission. (2020). White paper on artificial intelligence: A European approach to excellence and trust. https://commission.europa.eu/publications/white-paper-artificial-intelligence-european-approach-excellence-and-trust_en. (Accessed 13 November 2023).

European Commission. (2024). The act texts | EU artificial intelligence act. https://artificialintelligenceact.eu/the-act/. (Accessed 17 February 2025).

European Parliament. (2023). EU AI act: First regulation on artificial intelligence. https://www.europarl.europa.eu/news/en/headlines/society/20230601STO93804/eu-ai-act-first-regulation-on-artificial-intelligence. (Accessed 13 November 2023).

European Union. (2023). CE marking. https://europa.eu/youreurope/business/product-requirements/labels-markings/ce-marking/index_en.htm. (Accessed 5 March 2024).

European Union. (2024). Conformity assessment. https://europa.eu/youreurope/business/product-requirements/compliance/conformity-assessment/index_en.htm. (Accessed 17 February 2025).

Fan, W., Wang, S., Gu, X., Zhou, Z. Q., Zhao, Y., & Huo, W. D. (2021). Evolutionary game analysis on industrial pollution control of local government in China. *Journal of Environmental Management, 298*, Article 113499. https://doi.org/10.1016/j.jenvman.2021.113499

Fei, N., Lu, Z., Gao, Y., Yang, G., Huo, Y., Wen, J., Lu, H., Song, R., Gao, X., Xiang, T., Sun, H., & Wen, J.-R. (2022). Towards artificial general intelligence via a multimodal foundation model. *Nature Communications, 13*(1), 3094. https://doi.org/10.1038/s41467-022-30761-2

Floridi, L. (2019). Establishing the rules for building trustworthy AI. *Nature Machine Intelligence, 1*, 261–262. https://doi.org/10.1038/s42256-019-0055-y

Floridi, L. (2021). The end of an era: From self-regulation to hard law for the digital industry. *Philosophy & Technology, 34*(4), 619–622. https://doi.org/10.1007/s13347-021-00493-0

Friedman, D. (1991). Evolutionary games in economics. *Econometrica, 59*(3), 637–666. https://doi.org/10.2307/2938222

Gawer, A., & Cusumano, M. A. (2014). Industry platforms and ecosystem innovation. *Journal of Product Innovation Management, 31*(3), 417–433. https://doi.org/10.1111/jpim.12105

Gintis, H. (2009). In *Game theory evolving: A problem-centered introduction to modeling strategic interaction* (2nd ed.). Princeton University Press.

Gomes, L. A. D., Facin, A. L. F., Salerno, M. S., & Ikenami, R. K. (2018). Unpacking the innovation ecosystem construct: Evolution, gaps and trends. *Technological Forecasting and Social Change, 136*, 30–48. https://doi.org/10.1016/j.techfore.2016.11.009

Hacker, P., Engel, A., & Mauer, M. (2023). Regulating ChatGPT and other large generative AI models. In *Proceedings of the 2023 ACM conference on fairness, accountability, and transparency* (pp. 1112–1123). Chicago, IL, USA: Association for Computing Machinery. https://doi.org/10.1145/3593013.3594067.

Han, T. A., Lenaerts, T., Santos, F. C., & Pereira, L. M. (2022). Voluntary safety commitments provide an escape from over-regulation in AI development. *Technology in Society, 68*, Article 101843. https://doi.org/10.1016/j.techsoc.2021.101843

Han, T. A., Pereira, L. M., Lenaerts, T., & Santos, F. C. (2021). Mediating artificial intelligence developments through negative and positive incentives. *PLoS One, 16*(1), Article e0244592. https://doi.org/10.1371/journal.pone.0244592

Hannigan, T. R., Briggs, A. R., Valadao, R., Seidel, M.-D. L., & Jennings, P. D. (2022). A new tool for policymakers: Mapping cultural possibilities in an emerging AI entrepreneurial ecosystem. *Research Policy, 51*(9), Article 104315. https://doi.org/10.1016/j.respol.2021.104315

Heng, S., Tsilionis, K., Scharff, C., & Wautelet, Y. (2022). Understanding AI ecosystems in the Global South: The cases of Senegal and Cambodia. *International Journal of Information Management, 64*, Article 102454. https://doi.org/10.1016/j.ijinfomgt.2021.102454

The High-Level Expert Group on Artificial Intelligence (AI HLEG) of European Commission, the assessment list for trustworthy artificial intelligence (ALTAI) for self assessment. https://digital-strategy.ec.europa.eu/en/library/assessment-list-trustworthy-artificial-intelligence-altai-self-assessment, (2020)–. (Accessed 13 November 2023).

Jacobides, M. G., Cennamo, C., & Gawer, A. (2018). Towards a theory of ecosystems. *Strategic Management Journal, 39*(8), 2255–2276. https://doi.org/10.1002/smj.2904

Ji, S. F., Zhao, D., & Luo, R. J. (2019). Evolutionary game analysis on local governments and manufacturers' behavioral strategies: Impact of phasing out subsidies for new energy vehicles. *Energy, 189*, Article 116064. https://doi.org/10.1016/j.energy.2019.116064

Kaur, D., Uslu, S., Rittichier, K. J., & Durresi, A. (2022). Trustworthy artificial intelligence: A review. *ACM Computing Surveys, 55*(2), 39. https://doi.org/10.1145/3491209

Knott, A., Pedreschi, D., Jitsuzumi, T., Leavy, S., Eyers, D., Chakraborti, T., Trotman, A., Sundareswaran, S., Baeza-Yates, R., Biecek, P., Weller, A., Teal, P. D., Basu, S., Haklidir, M., Morini, V., Russell, S., & Bengio, Y. (2024). AI content detection in the emerging information ecosystem: New obligations for media and tech companies. *Ethics and Information Technology, 26*, 63. https://doi.org/10.1007/s10676-024-09795-1

Kunz, W. H., & Wirtz, J. (2024). Corporate digital responsibility (CDR) in the age of AI: Implications for interactive marketing. *Journal of Research in Interactive Marketing, 18*(1), 31–37. https://doi.org/10.1108/JRIM-06-2023-0176

Küspert, S., Moës, N., & Dunlop, C. (2023). The value chain of general-purpose AI: A closer look at the implications of API and open-source accessible GPAI for the EU AI act. https://www.adalovelaceinstitute.org/blog/value-chain-general-purpose-ai/. (Accessed 13 November 2023).

Laux, J., Wachter, S., & Mittelstadt, B. (2024). Trustworthy artificial intelligence and the European Union AI act: On the conflation of trustworthiness and acceptability of risk. *Regulation & Governance, 18*(1), 3–32. https://doi.org/10.1111/rego.12512

Li, B., Qi, P., Liu, B., Di, S., Liu, J., Pei, J., Yi, J., & Zhou, B. (2023). Trustworthy AI: From principles to practices. *ACM Computing Surveys, 55*(9), 177. https://doi.org/10.1145/3555803

Li, X., Tian, Y., Ye, P., Duan, H., & Wang, F. Y. (2023). A novel scenarios engineering methodology for foundation models in metaverse. *IEEE Transactions on Systems, Man, and Cybernetics: Systems, 53*(4), 2148–2159. https://ieeexplore.ieee.org/document/9999152.

Liu, H., Wang, Y., Fan, W., Liu, X., Li, Y., Jain, S., Liu, Y., Jain, A., & Tang, J. (2022). Trustworthy AI: A computational perspective. *ACM Trans. Intell. Syst. Technol., 14*(1), 4. https://doi.org/10.1145/354687

Lyapunov, A. M. (1992). The general problem of the stability of motion. *International Journal of Control, 55*(3), 531–534. https://doi.org/10.1080/00207179208934253

Madiega, T. (2023). General-purpose artificial intelligence - European Parliament. https://www.europarl.europa.eu/RegData/etudes/ATAG/2023/745708/EPRS_ATA(2023)745708_EN.pdf. (Accessed 13 November 2023).

Mazurek, G., & Małagocka, K. (2019). Perception of privacy and data protection in the context of the development of artificial intelligence. *Journal of Management Analytics, 6*(4), 344–364. https://doi.org/10.1080/23270012.2019.1671243

Morley, J., Elhalal, A., Garcia, F., Kinsey, L., Mökander, J., & Floridi, L. (2021). Ethics as a service: A pragmatic operationalisation of AI ethics. *Minds and Machines, 31*(2), 239–256. https://doi.org/10.1007/s11023-021-09563-w

Nannini, L., Alonso-Moral, J. M., Catalá, A., Lama, M., & Barro, S. (2024). Operationalizing explainable artificial intelligence in the European Union regulatory ecosystem. *IEEE Intelligent Systems, 39*(4), 37–48. https://ieeexplore.ieee.org/document/10489989.

OECD. (2019). Recommendation of the council on artificial intelligence. https://legalinstruments.oecd.org/en/instruments/OECD-LEGAL-0449#relatedInstruments. (Accessed 5 March 2024).

Papernot, N., McDaniel, P., Goodfellow, I., Jha, S., Celik, Z. B., & Swami, A. (2017). Practical black-box attacks against machine learning. In *Proceedings of the 2017 ACM on Asia conference on computer and communications security* (pp. 506–519). Abu Dhabi, United Arab Emirates: Association for Computing Machinery. https://doi.org/10.1145/3052973.3053009.

Powers, S. T., Linnyk, O., Guckert, M., Hannig, J., Pitt, J., Urquhart, N., Ekárt, A., Gumpfer, N., Han, T. A., Lewis, P. R., Marsh, S., & Weber, T. (2023). The stuff we swim in: Regulation alone will not lead to justifiable trust in AI. *IEEE Technology and Society Magazine, 42*(4), 95–106. https://ieeexplore.ieee.org/document/10410106.

Smith, J. M. (2012). *Evolution and the theory of games*. Cambridge University Press.

Stahl, B. C. (2022). Responsible innovation ecosystems: Ethical implications of the application of the ecosystem concept to artificial intelligence. *International Journal of Information Management, 62*, Article 102441. https://doi.org/10.1016/j.ijinfomgt.2021.102441

Stahl, B. C. (2023). Embedding responsibility in intelligent systems: From AI ethics to responsible AI ecosystems. *Scientific Reports, 13*(1), 7586. https://doi.org/10.1038/s41598-023-34622-w

Stix, C. (2021). Actionable principles for artificial intelligence policy: Three pathways. *Science and Engineering Ethics, 27*(1), 15. https://doi.org/10.1007/s11948-020-00277-3

Sun, Y., Wu, Z., Lan, J., Li, Y., & Dou, Z. (2024). Spatiotemporal distribution and dynamics evolution of artificial intelligence development in China. *Heliyon, 10*(1), Article e23885. https://doi.org/10.1016/j.heliyon.2023.e23885

Tanimoto, J. (2015). *Fundamentals of evolutionary game theory and its applications*. Tokyo: Springer Tokyo.

Thiebes, S., Lins, S., & Sunyaev, A. (2021). Trustworthy artificial intelligence. *Electronic Markets, 31*(2), 447–464. https://doi.org/10.1007/s12525-020-00441-4

Tramèr, F., Zhang, F., Juels, A., Reiter, M. K., & Ristenpart, T. (2016). *Stealing machine learning models via prediction APIs, 25th USENIX Security Symposium*. Austin, TX: USENIX Association. https://dl.acm.org/doi/10.5555/3241094.3241142.

Veale, M., & Borgesius, F. J. Z. (2021). Demystifying the Draft EU Artificial Intelligence Act — analysing the good, the bad, and the unclear elements of the proposed approach. *Computer Law Review International, 22*, 97–112. https://doi.org/10.9785/cri-2021-220402

Weibull, J. (1995). *Evolutionary game theory*. Cambridge: MA: The M.I.T. Press.

Wirtz, J., Kunz, W. H., Hartley, N., & Tarbit, J. (2023). Corporate digital responsibility in service firms and their ecosystems. *Journal of Service Research, 26*(2), 173–190. https://doi.org/10.1177/10946705221130467

Zhou, C., Li, Q., Li, C., Yu, J., Liu, Y., Wang, G., Zhang, K., Ji, C., Yan, Q., He, L., Peng, H., Li, J., Wu, J., Liu, Z., Xie, P., Xiong, C., Pei, J., Yu, P. S., & Sun, L. (2024). A comprehensive survey on pretrained foundation models: A history from BERT to ChatGPT. *International Journal of Machine Learning and Cybernetics*. https://doi.org/10.1007/s13042-024-02443-6